

Fifty-six conjectures in mathematics: the statements

1 Notation

p, q denote primes; γ is Euler’s constant, $e^\gamma = 1.781072\dots$; φ is Euler’s totient; $\phi = (1 + \sqrt{5})/2$; F_n the Fibonacci numbers; $p\#$ the primorial. For irreducible $f_1, \dots, f_k \in \mathbb{Z}[x]$ with positive leading coefficients, the Bateman–Horn singular series is

$$C(f_1, \dots, f_k) = \prod_p \frac{1 - \omega(p)/p}{(1 - 1/p)^k}, \quad \omega(p) = \#\{n \bmod p : p \mid f_1(n) \cdots f_k(n)\},$$

and the corresponding main term is

$$I(N) = \int_2^N \frac{dt}{\prod_i \log f_i(t)},$$

which absorbs the usual $1/\prod_i \deg f_i$. The Bateman–Horn conjecture [2] asserts $\#\{n \leq N : \text{all } f_i(n) \text{ prime}\} \sim C \cdot I(N)$ whenever $C \neq 0$; each Bateman–Horn conjecture below is an instance with its constant evaluated. For nonlinear systems the product converges only conditionally; throughout, it is taken in the canonical ordering of increasing primes, in which convergence follows from the equidistribution of $\omega(p)$ about its Chebotarev mean. Our numerical truncation wobble is an error bar for this canonically ordered product. We write $\text{Li}_k(x) = \int_2^x (\log t)^{-k} dt$. The twin singular series is $\mathfrak{S}(d) = 2C_2 \prod_{p \mid d, p > 2} \frac{p-1}{p-2}$ for even d , with $2C_2 = 1.3203236\dots$; the Goldbach series $\mathfrak{S}(n)$ is the same product over odd $p \mid n$.

Eight of the conjectures below are built over the Bateman–Horn move—choose an admissible system, compute its singular series, state the count—but none is a bare instance of it: 10 and 15 are family laws with derived constant distributions, 23 and 25 carry structural classifications, 24 is a searched chain, and 18, 19, 14 are contamination-calculus predictions (two fresh races and the negative control) whose mechanism is the contamination calculus of Conjecture 1(v). Each carries its classical calibration instance as an attributed remark. The heuristic is uniform, so we give it once. Model the primality of the values $f_i(n)$ as independent events of probability $1/\log f_i(n)$, corrected by the factor $(1 - \omega(p)/p)/(1 - 1/p)^k$ at each prime p to account for the joint local behaviour; multiplying the corrections and integrating gives $C \cdot I(N)$.

For locally integrable F we write $\mathcal{M}_x(F) = \frac{1}{\log x} \int_2^x F(t) \frac{dt}{t}$ for the logarithmic mean; all race conjectures below state their drift and null clauses through $\mathcal{M}_x$, so that “no persistent leader” and “systematic surplus” are claims about a defined functional rather than about pointwise behaviour. Drift clauses are stated at the *drift scale*—for a pair race, $\mathcal{M}_x(D(t) \log^2 t / \sqrt{t})$ converging to an explicit constant—never as $\mathcal{M}_x((D - T)/\sqrt{\pi}) \rightarrow 0$, which is vacuous for the drift coefficient (both $T/\sqrt{\pi}$ and its multiples have logarithmic mean $\rightarrow 0$). Convergence of $\mathcal{M}_x$ at drift scale is itself part of each such conjecture: a pure random-walk fluctuation model would make these means

divergent in variance, so their convergence encodes the Rubinstein–Sarnak-type almost-periodicity that the zero-oscillation hypothesis names.

Admissibility ($\omega(p) < p$ for all p) is checked for every constant computed. The natural-looking pair $\{n^2 + n + 1, n^2 + n + 3\}$ has $\omega(3) = 3$ and is rejected; the $k = 4$ branch of Conjecture 25 and the naive iterated chain behind Conjecture 24 are inadmissible for the same reason, and both exclusions are part of the statements.

Part I

Conjectures from a local–global random model

2 Principal conjectures

Lemma 1 (Orientation elimination and class assignment). *In $\psi_2(x) = \sum_{n \leq x} \Lambda(n)\Lambda(n+2)$, the total contribution of terms in which n or $n+2$ is a proper prime power is, apart from $O(x^{1/3} \log^2 x)$ from cubes and higher powers, carried entirely by the patterns $(q^2 - 2, q^2)$ with q prime and $q^2 - 2$ prime: the mirror pattern $(q^2, q^2 + 2)$ is annihilated by $3 \mid q^2 + 2$ for every prime $q > 3$. Moreover, for every prime $q \neq 5$, $q^2 - 2 \equiv 2$ or $4 \pmod{5}$ (never 1), and for every odd q , $q^2 - 2 \equiv 7 \pmod{8}$. (Two bounded exceptions: at $q = 5$ the term $(23, 25)$ lands outside the twin-start classes altogether, and at $q = 3$ the mirror term $(9, 11)$ survives, the annihilation beginning only at $q > 3$; each is a single term, absorbed in the bounded error.)*

Conjecture 1 (Twin races modulo 5 and 8, and the prime-power contamination calculus; apparently new). *Twin starts $p > 5$ lie in classes $p \equiv 1, 2, 4 \pmod{5}$. Let $D_1(x) = \pi_t(x; 5, 1) - \frac{1}{2}(\pi_t(x; 5, 2) + \pi_t(x; 5, 4))$ and*

$$T(x) = \frac{1}{2 \log^2 x} \sum_{\substack{q \leq \sqrt{x} \\ q^2 - 2 \text{ prime}}} \log q \log(q^2 - 2),$$

Then: (i) [drift law, at drift-scale normalization] the clause asserts two sub-claims, labelled separately because they can fail independently. (i-a) [mechanism] class 1 carries the systematic surplus T : one has the decomposition $D_1(t) = T(t) + R(t)$ in which the remainder R carries no deterministic component at the drift scale, so that the entire drift-scale deterministic content of D_1 is the contamination term and its coefficient is thereby identified. (i-b) [averaging] at the drift-scale normalization the remainder averages away:

$$\mathcal{M}_x \left(\frac{R(t) \log^2 t}{\sqrt{t}} \right) \longrightarrow 0, \quad \text{equivalently} \quad \mathcal{M}_x \left(\frac{D_1(t) \log^2 t}{\sqrt{t}} \right) \longrightarrow c_T,$$

where

$$c_T := \lim_{x \rightarrow \infty} \frac{T(x) \log^2 x}{\sqrt{x}} = \frac{1}{2} C(x, x^2 - 2)$$

is the half of the Bateman–Horn constant of the pair $(q, q^2 - 2)$ (the weighted census sum is $\sim C\sqrt{x}$, so the limit exists and is explicit). A normalization at the noise scale, $\mathcal{M}_x((D_1 - T)/\sqrt{\pi_t}) \rightarrow 0$,

would be vacuous for the coefficient of (i-a): since $T/\sqrt{\pi_t} \asymp 1/\log t$, one has $\mathcal{M}_x(T/\sqrt{\pi_t}) \asymp \log \log x / \log x \rightarrow 0$, so the condition holds with T replaced by 0, $2T$, or any fixed multiple, and does not express “exactly the removed mass.” The statement therefore normalizes at the drift scale, where the claimed limit is a nonzero constant. Convergence at this normalization is a Rubinstein–Sarnak-type regularity assertion: under a pure random-walk model the logarithmic mean would not converge (its variance grows like $\log x$), so (i-b) asserts almost-periodic structure for the twin race and is the precise content of the zero-oscillation hypothesis, while (i-a) is what fixes the limit at c_T rather than 0 or $2c_T$, and is the calculus. The convergence asserted in (i-b) selects the rigid branch of the dichotomy registered at Conjecture 9(iii); the two statements stand or fall together at drift scale; (ii) [zero deterministic drift in the symmetric differences] the symmetric classes carry no deterministic term at the same normalization: $\mathcal{M}_x((\pi_t(\cdot; 5, 2) - \pi_t(\cdot; 5, 4)) \log^2 t / \sqrt{t}) \rightarrow 0$, and likewise for each of the three pairwise differences among classes $\{1, 3, 5\} \bmod 8$ in clause (iv)—now a statement distinct from (i), since at drift scale “zero” and “ c_T ” are different limits; (iii) [sign density, conditional] conditional on the Gaussian fluctuation model for the normalized remainder in (i)–(ii), the logarithmic density of $\{D_1 > 0\}$ exists and equals $\frac{1}{2}$, approached from above with a finite- x excess of order $\log \log x / \log x$: the pointwise drift-to-noise ratio is $T/\sqrt{\pi_t} \asymp 1/\log t$, and its logarithmic average $\mathcal{M}_x(1/\log t) \sim \log \log x / \log x$ carries an extra $\log \log$ (the scale is not $1/\log x$). In contrast to Chebyshev’s races, the twin race has no persistent leader; (iv) [mod-8 companion] since $q^2 - 2 \equiv 7 \pmod{8}$ always (Lemma 1), the entire deterministic term falls on the single class $7 \pmod{8}$: with $D_7(x) = \frac{1}{3} \sum_{a \in \{1, 3, 5\}} \pi_t(x; 8, a) - \pi_t(x; 8, 7)$,

$$\mathcal{M}_x\left(\frac{D_7(t) \log^2 t}{\sqrt{t}}\right) \rightarrow 2c_T$$

(twice the mod-5 constant, undiluted—at this normalization the factor 2 is a falsifiable prediction, the mechanism and averaging claims separating here exactly as in (i-a)–(i-b)), while classes 1, 3, 5 are mutually symmetric. The mod-5 and mod-8 races are two projections of one mechanism, so their joint behaviour is a family test in the sense of the shape-of-agreement criterion; (v) [contamination calculus for balanced races] for a pattern $(n, n + d)$ and modulus m , call the race balanced if the prime-prime singular series is identical across the competing residue classes (as CRT gives for the classes compared in (i)–(iv)) and no mechanism other than prime powers contributes at the $\sqrt{x}$ scale under the zero-oscillation hypothesis. For balanced races the deterministic drift vector is computed by one operator. Define the contamination operator

$$\mathcal{C}_{d,m}(a; x) = \sum_{\text{orientations } o \in \{(q^2-d, q^2), (q^2, q^2+d)\}} \sum_{\substack{q \leq \sqrt{x}, \text{ } o \text{ prime-compatible} \\ \text{start}(o) \equiv a \pmod{m}}} \Lambda(q^2) \Lambda(\text{prime member of } o),$$

where prime-compatible means the non-square member is prime and no fixed prime annihilates the orientation. The provable layer is the orientation census and class assignment (which o survive, and where their starts land—Lemma 1 and its analogues); the conjectural layer is the transfer: the unweighted class counts are depressed by $\mathcal{C}_{d,m}(a; x) / \log^2 x$ relative to the clean classes, in the drift-scale limiting-log-mean sense of (i), with the mechanism and averaging claims separated as in (i-a)–(i-b)—for each class a , the deficit obeys $\mathcal{M}_x(\text{deficit}_a(t) \log^2 t / \sqrt{t}) \rightarrow c(a) := \lim \mathcal{C}_{d,m}(a; x) / \sqrt{x}$, an explicit Bateman–Horn constant per class, so every coefficient in the vector is identified—i.e. “exactly the removed mass,” which requires the weighted identity, partial summation uniform in classes, higher-power bounds (exponents $r \geq 3$ contribute $O(x^{1/3+\epsilon})$ against the operator’s $\asymp \sqrt{x}$ and are negligible—the one easy layer of the transfer), and the zero-oscillation hypothesis, and

is the calculus's single analytic conjecture rather than a rule. The drift is $\asymp \sqrt{x}/\log^2 x$ against a count fluctuation $\asymp \sqrt{x}/\log x$, i.e. smaller by $1/\log x$ in standard-deviation units at every height—so the calculus predicts occupation biases under long logarithmic aggregation, never a fixed nonzero standardized mean, and every verification in this family is designed around that fact. (Races with unequal primary constants, or with interfering orientations beyond those enumerated, are outside the calculus until those terms are separated.) In particular, for the cousin pattern $d = 4$ the calculus makes two predictions with no free choices: the orientation $(q^2 - 4, q^2)$ is dead apart from a single term ($q^2 - 4 = (q - 2)(q + 2)$ is composite for $q > 3$, while at $q = 3$ the value 5 is prime: one bounded term, whose start 5 lies outside the mod-5 cousin classes and contributes one bounded term to class $5 \bmod 8$), the orientation $(q^2, q^2 + 4)$ survives (no prime annihilates it), $q^2 + 4$ prime forces $q \equiv \pm 2 \pmod{5}$ for $q \neq 5$ (at $q = 5$ the value 29 is prime, but the start $25 \equiv 0 \pmod{5}$ lies outside the cousin start classes $\{2, 3, 4\}$), so the contaminated cousin-start class is $4 \pmod{5}$ (of $\{2, 3, 4\}$) and—since $q^2 \equiv 1 \pmod{8}$ —the class $1 \pmod{8}$ (of $\{1, 3, 5, 7\}$): both deficits converge, in the drift-scale limiting-log-mean sense of (i)–(ii), to $c^c = \lim_{x \rightarrow \infty} T^c(x) \log^2 x / \sqrt{x}$, the Bateman–Horn constant of the pair $(q, q^2 + 4)$, where $T^c(x) = \frac{1}{\log^2 x} \sum_{q \leq \sqrt{x}, q \neq 5, q^2 + 4 \text{ prime}} \log q \log(q^2 + 4)$.

Conjecture 2 (The pair-level Montgomery–Soundararajan reduction; apparently new). *For the window field of Conjecture 8(ii), the diagonal variance of the twin-pair count reduces to a pinned singular-series average. With $R(h) = C_4(0, 2, h, h + 2)/\mathfrak{S}(2)^2$ for $h \geq 1$ (zero when the offset set is degenerate or inadmissible—all odd h , and $h = 2$ since $\{0, 2, 4\}$ dies at 3; consecutive twin pairs cannot overlap beyond $(3, 5, 7)$, so no overlap term exists) and*

$$G(H) = \sum_{1 \leq h \leq H} \left(1 - \frac{h}{H}\right) (R(h) - 1),$$

the derived identity is

$$\frac{\text{Var}_t \pi_2(t; H)}{\mathbb{E}_t \pi_2(t; H)} = 1 + \frac{\mathfrak{S}(2) (2G(H) - 1)}{\log^2 x} + o\left(\frac{1}{\log^2 x}\right) :$$

with $p = \mathfrak{S}(2)/\log^2 x$ the per-site intensity, the variance is $Hp(1 - p) + 2p^2 HG(H)$, so $\text{Var}/\mathbb{E} = 1 - p + 2pG(H)$ —the Bernoulli diagonal term $-p = -\mathfrak{S}(2)/\log^2 x$ is of exactly the order of the claimed error and cannot be dropped (it is the same diagonal that, in the single-prime case, merges into Montgomery–Soundararajan's constant $\gamma + \log 2\pi - 1$). With that term retained, the entire deficit left open at Conjecture 8(ii) is the single computable function G plus the explicit Bernoulli correction—the pair analogue of Montgomery–Soundararajan's reduction of the prime-count variance to $\sum_{h \leq H} \mathfrak{S}(h)(H - h)$ [19]. Clause (i) is a conditional proposition, not an independent conjecture: given quantitative Hardy–Littlewood for 4-tuples, the reduction follows. The conjectural clause is (ii): $-G(H)/\log H$ diverges—the pinned average grows strictly faster than the single-prime secondary term $\frac{1}{2}(\log H + \gamma + \log 2\pi - 1)$, so pair counts in windows are more sub-Poisson than prime counts—with the caveat that data to $H = 3000$ cannot distinguish divergence from a pure log with a large constant (nor $c \log H \log \log H$ from $(\log H)^\alpha$): what the computation establishes is only that the local slope exceeds the single-prime value sevenfold across the computable range; the divergence and its exact form are registered, not evidenced beyond that.

Conjecture 3 (Least primes in progressions; part (i) previously stated [20, 21, 30, 29], part (ii) apparently new). *Let $p(a, q)$ be the least prime $\equiv a \pmod{q}$ and $U(a, q) = \text{Li}(p(a, q))/\varphi(q)$.*

Then: (i) $U \Rightarrow \text{Exp}(1)$ marginally as $q \rightarrow \infty$; and, under the additional hypothesis that the class processes are asymptotically jointly independent in the extreme-value sense (which marginal convergence alone does not supply), $\max_a U - H_{\varphi(q)}$ converges to a Gumbel law ($H_n = \sum_{k \leq n} 1/k$), the order-statistics form of Wagstaff's $\max_a p(a, q) \sim \varphi(q) \log^2 q$; (ii) [canonical deficit, dyadic-average form] define, with no reference to any control model, $\Theta(q) = (1 - \mathbb{E}_a[U(a, q)]) \log q$. The conjecture is stated for dyadic averages over prime moduli—pointwise convergence along every prime q may be too strong, since arithmetic fluctuations in q could persist:

$$\frac{1}{\#\{q \in [Q, 2Q] \text{ prime}\}} \sum_{\substack{Q \leq q \leq 2Q \\ q \text{ prime}}} \Theta(q) \rightarrow \Theta > 0 \quad (Q \rightarrow \infty).$$

(The formal claim is model-free: existence and strict positivity of the dyadic-average limit. The stronger reading—that Θ exceeds the discreteness level any parity-aware control produces—is the diagnostic layer below, not part of the display, since a control constant has no place in a model-free statement.) The limiting distribution of $\Theta(q)$ over the dyadic block is left open, and in particular we do not assert $\text{sd}(\Theta(q)) \rightarrow 0$: fluctuations driven by the arithmetic of q and of $q - 1$ (through the singular-series pair terms $\mathfrak{S}(kq)$ and the class structure mod q) may well persist at bounded size, so $\Theta(q)$ need not concentrate around its dyadic mean.

Conjecture 4 (The Goldbach lane race; apparently new). For $n \equiv 2 \pmod{4}$ let $R_1(n)$, $R_3(n)$ be the ordered Goldbach representation counts in the lanes $p \equiv q \equiv 1$ and $p \equiv q \equiv 3 \pmod{4}$ (Conjecture 5), and $D(n) = R_3(n) - R_1(n)$. Prime-square terms $n = q^2 + p$ (q odd prime) land exclusively in the (1, 1) lane, since $q^2 \equiv 1 \pmod{4}$ and then $n - q^2 \equiv 1 \pmod{4}$ automatically. Consequently the (3, 3) lane leads on average: with

$$D_{\text{sys}}(n) = \frac{2}{\bar{\ell}(n)} \sum_{\substack{q \text{ odd prime}, q \leq \sqrt{n} \\ n - q^2 \text{ prime}}} \log q \log(n - q^2), \quad \bar{\ell}(n) = \frac{n - 6}{\int_3^{n-3} \frac{dt}{\log t \log(n - t)}},$$

(the analytic mean of $\log p \log(n - p)$ under the lane profile; the empirical lane mean is its estimator). Write $\mathbb{E}_x$ for the average over n drawn from $\{n \equiv 2 \pmod{4}, n \leq x\}$ with weight proportional to $1/n$ (the logarithmic sampling measure, now explicit). Then: (i) [drift law] $\mathbb{E}_x[D - D_{\text{sys}}] = o(\mathbb{E}_x[D_{\text{sys}}])$ as $x \rightarrow \infty$: the (3, 3) lane leads on average by exactly the square contamination; (ii) [weighted mean and internal null lane] $\mathbb{E}_x[D_{\text{sys}}]$ is given by the Hardy–Littlewood (Conjecture H) local model for $n - q^2$, whose local factors vary strongly with n —most drastically at $p = 3$: for $n \equiv 1 \pmod{3}$, $3 \mid n - q^2$ for every prime $q > 3$, so the contamination collapses to a single boundary family, the $q = 3$ term (when $n - 9$ is prime): the complementary possibility $n - q^2 = 3$ is empty on this ensemble, since $n \equiv 2 \pmod{4}$ forces $n - 3 \equiv 3 \pmod{4}$, never a square. The family is negligible in logarithmic mean, so the race carries no drift at the systematic scale on the subprogression $n \equiv 10 \pmod{12}$: with the positive comparison scale $A_G(x) := \mathbb{E}_x^{(\neq 1 \pmod{3})}[D_{\text{sys}}]$, the logarithmic-ensemble mean of the systematic term over the contaminated lane $n \not\equiv 1 \pmod{3}$ (a comparison against this subprogression's own systematic term, which vanishes, would be empty), the claim is $\mathbb{E}_x[D] = o(A_G(x))$ on $n \equiv 10 \pmod{12}$, while the drift concentrates on $n \not\equiv 1 \pmod{3}$: an internal null lane, predicted by the same mechanism that predicts the lead; (iii) [sign density] the per- n drift-to-noise ratio $\kappa(n) = D_{\text{sys}}(n)/\sqrt{\mathfrak{S}(n)J(n)}$ (where $J(n) = \int_3^{n-3} dt/(\log t \log(n - t))$), so that $\mathfrak{S}(n)J(n)$ is the total ordered representation count) satisfies $\kappa(n) \asymp c(n)/\log n \rightarrow 0$; under the Gaussian fluctuation model the logarithmic

density of $\{D > 0\}$ is therefore $\frac{1}{2}$ in the limit, with the finite- x sign fraction predicted to be $\mathbb{E}_x[\Phi(\kappa)]$ (Φ the standard normal distribution function)—decaying, but far from $\frac{1}{2}$ at any accessible height.

Conjecture 5 (The least Goldbach summand: exponential law and ordering deficit; the time-changed law and deficit constant apparently unstated). *Let $s(n)$ be the least prime $p \nmid n$ with $n - p$ prime (the exclusion $p \nmid n$ removes a trivial obstruction: $p \mid n$ forces $p \mid n - p$, so $n - p$ is composite except in the diagonal case $n = 2p$, which would otherwise distort the hazard at one atypical point), and time-change by the expected-arrivals clock $U(n) = \mathfrak{S}(n) \sum_{p \leq s(n), p \nmid n} 1/\log(n - p)$. Then: The ensemble is fixed first, since $U(n)$ is deterministic at each n and an expectation therefore needs a declared measure: for a scale X ,*

$$\mathbb{E}_X U = \frac{\sum_{X < n \leq 2X, 2|n} U(n)/n}{\sum_{X < n \leq 2X, 2|n} 1/n}, \quad \Theta_G(X) = (1 - \mathbb{E}_X U) \log X.$$

(i) [limit law] under $\mathbb{E}_X$ -sampling, $U \Rightarrow \text{Exp}(1)$ as $X \rightarrow \infty$; (ii) [canonical ordering deficit] $\Theta_G(X) \rightarrow \Theta_G > 0$ —the Goldbach sibling of the least-prime deficit of Conjecture 3(ii), produced by the same occupancy mechanism: the ordered primes fill the available representations faster than an exchangeable model, so the first arrival comes early and $\mathbb{E}_X[U] < 1$.

Conjecture 6 ($n^2 + 2^n$; sequence is OEIS A064539). *Every prime of the form $n^2 + 2^n$ with $n > 1$ has $n \equiv 3 \pmod{6}$ (elementary), and: (i) infinitely many such primes exist, with counting function $\asymp \log N$; (ii) [candidate law, entanglement-aware form] for a finite set S of primes $p \geq 5$ (the primes 2 and 3 are exact in the lane reduction, $\delta_2 = \delta_3 = 0$, and $\text{ord}_2 2$ is undefined) let D_S be the exact joint density, within the lane $n \equiv 3 \pmod{6}$, of n with $p \nmid n^2 + 2^n$ for all $p \in S$, computed over the full joint period $\text{lcm}_{p \in S} \text{lcm}(6, p \cdot \text{ord}_p 2)$, and set $\kappa_S = 3 D_S / \prod_{p \in S} (1 - 1/p)$: the factor $3 = 2 \cdot \frac{3}{2}$ is the exact lane bonus at the primes 2 and 3 (within the lane $\delta_2 = \delta_3 = 0$, so each contributes its full $(1 - 1/p)^{-1}$), included in the definition so that κ_S is the complete local constant of the count in (ii-c). The conjecture is stated along the canonical exhaustion $S_z = \{p \text{ prime} : 5 \leq p \leq z\}$, and its three layers are logically separate and are asserted separately: (ii-a) the sequence κ_{S_z} converges as $z \rightarrow \infty$ (a general net over arbitrary finite S carries an order-of-exhaustion ambiguity we do not need), with limit κ ; (ii-b) whether the limit factors—i.e. whether the exact finite-level factorization observed below persists, so that entanglement is asymptotically absent—is an open question, expressly not counted among this paper’s conjectural assertions, on which (ii-a) takes no position; and (ii-c) the count is $\sim \sum_{n \leq N, n \equiv 3 \pmod{6}} \kappa / \log(n^2 + 2^n)$ —a genuine further step, since a convergent local density does not automatically govern the global count. The working value $\kappa^* = 4.2734\dots$ is a hybrid, not a computation of any single κ_{S_z} : the lane factor 3 times the CRT-exact core S_{19} times independent per-prime factors for $19 < p \leq 300$ —so $\kappa^* = \kappa_{S_{300}}$ exactly if and only if the observed exact factorization persists to 300.*

Conjecture 7 (Fermat quotients: the multibase subtorus law and the Wieferich ledger; single-base heuristic previously stated [22], the vertical multibase law apparently new). *With $q_p(a) = (a^{p-1} - 1)/p \pmod{p}$ (defined for primes $p \nmid a$; all clauses below range over such p only) and $q_p = q_p(2)$, five clauses: (0) [structure, classical] $q_p(ab) \equiv q_p(a) + q_p(b) \pmod{p}$ (the Eisenstein–Lerch homomorphism; verified exactly on every tested prime), so for multiplicatively dependent bases the vector $(q_p(a_1), \dots, q_p(a_r))/p$ is confined to the rational subtorus cut out by the relations; (iv) [multibase equidistribution, the claimed law] for multiplicatively independent bases the vector equidistributes on the full torus as p varies—the vertical joint law, complementing the fixed- p ,*

varying-base statistics of Ostafe–Shparlinski and Cobeli–Zaharescu—with correlations vanishing and the same LIL calibration as (i); (v) [simultaneous Wieferich] the expected count of $p \leq x$ with $q_p(2) = q_p(3) = 0$ has convergent sum $\sum 1/p^2$: at most finitely many simultaneous Wieferich primes exist (the single-base folklore of this accounting is Conrad’s), and the empirical list is empty to 10^7 . Three further clauses on the base-2 quotient alone: (i) [global equidistribution] q_p/p equidistributes on $[0, 1)$ with discrepancy at exactly the calibrated random-model scale: in its sharp form,

$$\limsup_{x \rightarrow \infty} \frac{\sqrt{\pi(x)} D_{\text{KS}}(x)}{\sqrt{\log \log \pi(x)}} = \frac{1}{\sqrt{2}},$$

the Chung–Smirnov law-of-the-iterated-logarithm constant for an i.i.d. uniform sequence—demanding both no more (a bound $O(\pi(x)^{-1/2})$ would be stronger than the random model supports, the same over-demand that killed the Mertens conjecture) and no less than the model earns: the upper bound alone would be consistent with hidden rigidity, and the matching lower bound asserts the quotients fluctuate like genuine noise; (ii) [shrinking targets] for every fixed $K \geq 1$, and uniformly for $K = K(x) \leq \log x$, $\#\{p \leq x : q_p < K\} \sim \sum_{p \leq x} \min(1, K/p)$ —the fixed- K case down to $K = 1$ is what governs the zero event and does NOT follow from (i), since $\{q_p = 0\}$ is a target of shrinking measure $1/p$; (iii) [Wieferich count, fluctuation-calibrated, with the probability space declared] $W(x) = \#\{\text{Wieferich } p \leq x\}$ is a deterministic function, so its fluctuation clauses are stated in the two legitimate forms, an unqualified CLT for a fixed sequence having no meaning: the deterministic envelope conjecture

$$\limsup_{x \rightarrow \infty} \frac{W(x) - \log \log x}{\sqrt{2 \log \log x \log \log \log x}} = 1$$

(the iterated-logarithm scale at variance $V(x) \sim \log \log x$; a bound $O_\varepsilon(V^{1/2+\varepsilon})$ would be weaker than the model’s exact prediction), and the almost-sure central limit law, at the correct logarithmic weighting: with $u = \log \log x$ the variance-time of the count, sample u with density du/u on $[e, L]$, $L = \log \log X$ (equivalently: $\log u = \log \log \log x$ uniform); then the sampled distribution of $(W(x) - u)/\sqrt{u}$ converges to $N(0, 1)$ as $X \rightarrow \infty$. The weighting is what makes the clause true, and the two natural alternatives both fail. A block-sampled version on $[X, X^A]$ fails because only $\log A = O(1)$ new events enter the block. A uniform-in- u version on $[1, L]$ fails too, by Brownian scaling: with $W(e^u) - u \approx B(u)$ and $u = Ls$, the self-similarity $B(Ls)/\sqrt{Ls} \stackrel{d}{=} \tilde{B}(s)/\sqrt{s}$ means the uniform- u empirical distribution converges not to Φ but to the random occupation functional $\int_0^1 \mathbf{1}\{\tilde{B}(s)/\sqrt{s} \leq z\} ds$ —values at proportional times stay correlated, so uniform sampling never averages over enough independent scales. The du/u weighting is the classical almost-sure-CLT weighting (the continuous analogue of the $1/k$ weights), for which the model does predict almost-sure convergence of the sampled law to Φ . Both clauses are conjectured on the Crandall–Dilcher–Pomerance side of the model choice that clause (iv)’s discussion records.

Conjecture 8 (Uniform quantitative de Polignac; the core is previously stated). (i) $\pi_d(x) = \#\{p \leq x - d : p, p + d \text{ prime}\} = \mathfrak{S}(d) \text{Li}_2(x) (1 + o(1))$ uniformly for even $d \leq (\log x)^A$. (ii) [residual field, with derived covariance kernel] The same primes participate in many pair counts, and the covariance of two of them is carried by triple configurations sharing one prime. For even $d \neq d'$ define the overlap constant

$$K(d, d') = C_3(0, d, d') + C_3(0, d', d + d') + C_3(0, d, d + d') + C_3(0, |d - d'|, \max(d, d')),$$

where C_3 denotes the Hardy–Littlewood triple constant of the offset set (degenerate or inadmissible triples contributing 0). The field is randomized canonically—covariances of the deterministic

cumulative counts are otherwise undefined: for t uniform on $[x, 2x]$ and a window length H , let $\pi_d(t; H) = \#\{t < n \leq t + H : n, n + d \text{ prime}\}$ and let $Z_d(t; H)$ be its studentization. The mean count per window is $\asymp H/\log^2 x$, and the regime is set by that ratio (a phase condition: a bare $H \rightarrow \infty$ does not fix the regime):

$$\text{Cov}_t(\pi_d, \pi_{d'}) \sim \frac{K(d, d') H}{\log^3 x}, \quad \rho(d, d') \sim \frac{K(d, d')}{\sqrt{\mathfrak{S}(d)\mathfrak{S}(d')} \log x},$$

valid for $\max(d, d') = o(H)$, which is the standing restriction of this clause and holds throughout the window class fixed below. In general each overlap orientation o —the in-window displacements $h_o \in \{0, d, -d', d - d'\}$ and their reflections—enters with multiplicity $(H - |h_o|)_+$ rather than H , so that the shared-prime term reads

$$\frac{1}{\log^3 x} \sum_o (H - |h_o|)_+ K_3^{(o)}(d, d'), \quad \sum_o K_3^{(o)}(d, d') = K(d, d'),$$

and an orientation with $|h_o| > H$ is absent from the window altogether; under $\max(d, d') = o(H)$ one has $(H - |h_o|)_+ = H(1 + o(1))$ for every orientation and the displayed form is recovered, so the uniform-in- d statement carries that restriction throughout. Here the shared-prime (triple) term is the first covariance contribution but not the whole of it: the full expansion is

$$\text{Cov}_t(\pi_d, \pi_{d'}) = \frac{K(d, d') H}{\log^3 x} + \frac{1}{\log^4 x} \sum_{|h| \leq H} (H - |h|) K_4(d, d'; h) + \text{error},$$

with K_4 the centred singular series of the off-index four-point sets $\{0, d, h, h + d'\}$ —the cross analogue of the pinned diagonal average of Conjecture 2—where degenerate offset relations are handled inside the kernels: whenever the four-point set collapses (repeated entries, as at $h = 0$, $h = d$, $h = -d'$, or $d' - d = \pm h$) the configuration is a triple or pair and its contribution belongs to the K -term (or to the mean), so K_4 is defined as the centred singular series of genuinely four-element sets, with special pairs such as $d' = 2d$ acquiring their extra overlap orientations in $K(d, d')$'s census rather than double-counted across kernels. The triple term dominates only where the pinned sum satisfies $\sum(1 - |h|/H)K_4 = o(\log x)$; the dominance range is stated as an explicit function class: for $H = \lambda(\log x)^\alpha$, any fixed $\lambda > 0$ and any $\alpha \geq 1$ with $\alpha > A$ —the inequality on the exponents that makes $\max(d, d') = o(H)$ hold uniformly for all $d, d' \leq (\log x)^A$ —the per-shift centred average grows only polylogarithmically in H (the Conjecture-2 scale), so $\sum(1 - |h|/H)K_4/H = O((\log \log x)^{2+\varepsilon}) = o(\log x)$ and the triple term dominates throughout the polylogarithmic window class; the conjecture asserts the two-term form on exactly that class and takes no position outside it. When $H/\log^2 x \rightarrow \lambda$ the window counts converge jointly to Poisson laws with these vanishing covariances, and when $H/\log^2 x \rightarrow \infty$ every fixed finite collection $\{Z_{d_1}, \dots, Z_{d_r}\}$ is asymptotically jointly normal with independent limiting coordinates: both kernels are finite- x corrections to independence (test below), not nondegenerate limiting kernels. Both distributional conclusions are asserted under an explicit hypothesis beyond the displayed covariances, namely uniform Hardy–Littlewood tuple estimates on the window class fixed above, strong enough to control all higher joint cumulants of the window counts: every joint cumulant of order at least three vanishes in the respective normalizations. The diagonal correction to $\text{Var}_t(\pi_d)$ is Conjecture 2's pinned average, stated there.

Conjecture 9 (The sub-diffusivity of balanced prime races; apparently new). *Model a balanced race (Conjecture 1(ii)'s symmetric class differences) as the walk of its class-assignment steps,*

the process defined at event index: enumerate the pattern occurrences $n_1 < n_2 < \dots$ in the two competing classes, set $\xi_i = \pm 1$ by class membership of n_i , and let $\rho_k(x)$ denote the lag- k sample autocorrelation of (ξ_i) over the events with $n_i \leq x$. Then the walk is measurably sub-diffusive: (i) [decaying-repulsion law] the step autocorrelations are negative at small lags—the race analogue of the Lemke Oliver–Soundararajan consecutive-pattern repulsion [24], which supplies the mechanism—and, since that repulsion is scale-dependent rather than fixed, the conjectured form is a decay law

$$\rho_k(x) = -\frac{c_k \log \log x + d_k}{\log x} (1 + o(1)),$$

with constants c_k, d_k in principle derivable from the LOS correlation series (underived here; and the measured values $\rho_1 \approx -0.037$, $\rho_2 \approx -0.009$ at 10^9 —universal across twin and cousin races mod 5 and 8 to within 0.004, each ~ 50 standard errors from zero—are observations at one height, consistent with $c_1 \approx 0.25$ there, not asymptotic constants); (ii) [long memory, measured] short lags do not exhaust the deficit: the running maxima $M(x) = \max_{t \leq x} |D(t)|/\sqrt{\text{count}}$ of the four races land at quantiles 0.02–0.17 of the simulated independent-step null (median 2.34 at this span), and still only 0.03–0.23 after correcting for $\rho_{1,2}$: the negative correlation persists across lags, so the cumulative diffusivity $\sigma^2(x) = 1 + 2 \sum_k \rho_k(x)$ sits well below 1 at this height. The model layer is declared: (ξ_i) is a deterministic, nonstationary family and the $\rho_k(x)$ are empirical statistics of it, so the Green–Kubo identity $\sigma^2 = 1 + 2 \sum_k \rho_k$ is invoked under an explicit local-stationarity model hypothesis—the step process restricted to a window $[x, 2x]$ is modelled as a stationary sequence whose autocorrelations are the measured $\rho_k(x)$ —and the variance statement can equivalently be taken as a direct definition of $\sigma^2(x)$ through block averages of $(S_{n+m} - S_n)^2/m$ over the events up to x , a form in which no stationarity is assumed at all. The Green–Kubo sum requires more than the fixed-lag law of (i): the per-lag decay controls each $\rho_k(x)$ but not the sum unless the decay is uniform in k with summable tail, since the number of relevant lags may grow with x . We therefore state the needed hypothesis as part of the clause: $\rho_k(x)$ obeys the law of (i) uniformly for $k \leq K(x)$ with $\sum_{k > K(x)} |\rho_k(x)| = o(\log \log x / \log x)$ for some $K(x) \rightarrow \infty$, and—since the tail bound alone leaves the growing head sum uncontrolled—the repulsion constants are summable, $\sum_k c_k < \infty$ and $\sum_k |d_k| < \infty$, so that $\sigma^2(x) = 1 - 2(\sum_k c_k \log \log x + \sum_k d_k) / \log x (1 + o(1))$ is a genuine consequence rather than an unjustified interchange of limits. The measured correlation length at 10^9 (a few tens of lags, with $|\rho_k|$ decreasing geometrically in the observable range) is consistent with both hypotheses. Under them, $\sigma^2(x)$ returns to 1 asymptotically, making sub-diffusivity a finite-height phenomenon of $\log \log x / \log x$ size, like every other second-order law in this paper; (iii) [registered dichotomy, open] asymptotically the two possibilities are mutually exclusive, and which of them holds is the open question. Either the fixed-lag expansion of (i) extends uniformly in k with summable constants—in which case the hypotheses of (ii) hold, (ii) applies, and the race is asymptotically diffusive with $\sigma^2(x) \rightarrow 1$ (running maxima $\sim \sqrt{2 \log \log x}$, Darling–Erdős class)—or that expansion fails at long lags, and only in that case can the race inherit the almost-periodic rigidity that Rubinstein–Sarnak-type structure [6] forces on classical races (maxima $\asymp (\log \log \log x)^A$, the Montgomery–Ng class). Rigidity therefore requires a failure of the hypotheses of (ii) at long lags, and cannot coexist with them. No explicit formula is known for twin patterns, the two branches differ only beyond any computable height, and we register the dichotomy without choosing in this conjecture. What is conjectured here is therefore the fixed-lag law (i) outright, and the variance law of (ii) conditionally on its stated uniformity and summability hypotheses, which belong to the finite-height model layer and are not asserted asymptotically: asserting them would silently select the diffusive branch of (iii). The pattern-specific selection

is made elsewhere—Conjecture 1 asserts the rigid branch for its races, through (i-b) for the contaminated combinations and through the drift-scale nulls of its clause (ii) for the symmetric differences—and consistency between the two conjectures then requires exactly that the hypotheses of (ii) fail at long lags for those races.

3 Family laws and second-order refinements

Conjecture 10 (Uniform quadratic de Polignac; family law apparently unstated, the $d = 2$ instance previously stated, cf. OEIS A080149, [32]). *For even $d \geq 2$ let $\pi_d^q(N) = \#\{n \leq N : n^2 + 1, n^2 + 1 + d \text{ both prime}\}$ and $C(d) = C(x^2 + 1, x^2 + 1 + d)$. Then: (i) [uniform family law] for every fixed $B > 0$, $\pi_d^q(N) = C(d) I_d(N)(1 + o(1))$ uniformly over even $d \leq (\log N)^B$; (ii) [registered rate, strictly stronger] there is $\eta_B > 0$ with*

$$\sup_{\substack{2 \leq d \leq (\log N)^B \\ 2|d}} \left| \frac{\pi_d^q(N)}{C(d) I_d(N)} - 1 \right| \ll_B (\log N)^{-\eta_B}$$

—an error clause that no version of Bateman–Horn supplies, registered separately from (i) since its analytic content is independent; (iii) [family statistic: the law of the constants] each local factor of $C(d)$ depends only on $d \bmod p$, and the factors at any finite set of primes are exactly independent as d varies (CRT), with the passage to the infinite product controlled by the convergent tail-variance sum—which supplies the uniform integrability (via L^2 -boundedness of the normalized partial products) that the moment identities require, since almost-sure convergence alone would not carry means through the limit; so the moments of $C(d)$ over even $d \leq D$ converge, as $D \rightarrow \infty$, to derived Euler products; in particular

$$\frac{1}{\#} \sum_{2|d \leq D} C(d) \rightarrow \bar{C} = 2.7456\dots, \quad \text{sd}(C(d)) \rightarrow 1.6840\dots,$$

and the empirical distribution of $C(d)$ converges to the law of the random product $\prod_p f_p(U_p)$ with independent uniform residues U_p . In particular ($d = 2$, the classical instance): $\#\{n \leq N : n^2 + 1, n^2 + 3 \text{ both prime}\} \sim C(2) I(N)$, $C(2) = 2.954014\dots$

Conjecture 11 (First occurrences of prime gaps, liminf form; the family is classical, the quantified form apparently unstated). *Let $\mathcal{G}$ be the set of even g that occur as gaps between consecutive primes (all even g , under Polignac’s conjecture), and for $g \in \mathcal{G}$ let $p(g)$ be the prime starting the first such gap. Then: (i) $\log p(g)/\sqrt{g}$ is bounded between positive constants on $\mathcal{G}$; (ii) [liminf law, stated as the dual of the maximal-gap limsup, conditional on the realization hypothesis below]*

$$\liminf_{\substack{g \rightarrow \infty \\ g \in \mathcal{G}}} \frac{\log p(g)}{\sqrt{g}} = \sqrt{e^\gamma/2} = 0.943682\dots$$

The inversion of Granville’s $\limsup G(X)/\log^2 X = 2e^{-\gamma}$ into first-occurrence coordinates is not automatic: the limsup controls how large maximal gaps get, not which gap sizes are realized as first occurrences near the envelope. Clause (ii) is therefore conditional on the realization hypothesis: infinitely many $g \in \mathcal{G}$ first occur inside a maximal-gap event at the Cramér–Granville envelope scale, i.e. with $g = (2e^{-\gamma} + o(1)) \log^2 p(g)$. Under that hypothesis the displayed value is forced by algebra; without it only the lower bound $\liminf \log p(g)/\sqrt{g} \geq \sqrt{e^\gamma/2}$ follows from

the envelope. The value is registered as that dual (against the slope-1 laws), not claimed as an independent phenomenon; (iii) [second-order, singular-series dependence] sampling g uniformly from the realized gaps $\mathcal{G} \cap [G, 2G]$ and letting $G \rightarrow \infty$,

$$\log p(g) = \sqrt{g} + \frac{1}{2} \log g - \frac{1}{2} \log \mathfrak{S}^*(g) + E_g, \quad \mathfrak{S}^*(g) = \prod_{\substack{p|g \\ p>2}} \frac{p-1}{p-2},$$

where the error E_g converges in distribution: for the sampled g ,

$$\Pr[E_g \leq t] \rightarrow 1 - \exp(-e^{2(t-\mu)}) \quad (G \rightarrow \infty)$$

for some centring constant μ —the min-type (reversed) Gumbel law at scale $\frac{1}{2}$, i.e. the law of $\frac{1}{2} \log W$ for W a unit exponential. The scale is forced by the hazard computation: the cumulative intensity of gap- g events is $\Lambda_g(y) \asymp (\mathfrak{S}(g)/y^2) \exp(y - g/y)$ in $y = \log x$ (the factor $e^{-g/y}$ is the no-intervening-prime probability), so $\frac{d}{dy} \log \Lambda_g = 1 + g/y^2 - 2/y = 2 + o(1)$ at the centring point $y_0 \approx \sqrt{g}$, and the first-passage identity $\Lambda_g(y) = W$ gives $E_g = y - y_0 = \frac{1}{2} \log W + o(1)$. The scale is $\frac{1}{2}$ and not 1 precisely because the slope of the log-intensity at y_0 is 2 (both the y -term and the g/y -term contribute), and the same slope-2 expansion is what produces the $\frac{1}{2}$ coefficients of the centring. (Gumbel-type modelling of gap extremes and first occurrences is Kourbatov–Wolf territory [15, 25]; the claim of this clause is only the $\mathfrak{S}^*$ -centring and the scale- $\frac{1}{2}$ first-passage form, not the extreme-value framework. Sampling uniformly over the realized gaps in $[G, 2G]$ conditions on realization, which the independent first-passage model does not represent; the clause conjectures that this selection does not shift the limiting law, and that assumption is part of what a refutation would refute.) Smooth gaps, whose tuple constant is large, appear earlier by exactly half a logarithm of that constant—the waiting-time first-passage computation makes the coefficient $-\frac{1}{2}$, not a free parameter, while μ is left as the one unresolved constant of the clause. (The restriction to $\mathcal{G}$ is needed: $p(g)$ is not known to be defined for every even g , and quantifying over all g would smuggle Polignac’s conjecture into the statement. The $\mathfrak{S}^*$ -clause records that the Hardy–Littlewood factor of g materially shifts waiting times; it does not move the liminf constant, since $\log \mathfrak{S}^*(g) = O(\log \log g)$.)

Conjecture 12 (The Stern lane race; apparently new). In Stern’s representation problem $n = p + 2k^2$ (n odd), race the k -even and k -odd lanes. Prime-square contamination of the Λ -weighted lane counts comes from $n = q^2 + 2k^2$ —the norm form $x^2 + 2y^2$ of $\mathbb{Q}(\sqrt{-2})$ with prime x —and since $q^2 \equiv 1 \pmod{8}$ while $2k^2 \equiv 0$ or $2 \pmod{8}$ by the parity of k : (i) [class assignment, direct congruence] contamination sits in the k -even lane iff $n \equiv 1 \pmod{8}$, in the k -odd lane iff $n \equiv 3 \pmod{8}$, and for $n \equiv 5, 7 \pmod{8}$ no square contamination exists at all: $q^2 + 2k^2 \equiv 1$ or $3 \pmod{8}$ always, a two-line congruence (genus theory enters only for the converse question of which n are represented), so the null classes are provable, not conjectured; (ii) [drift law, in the ensemble form of Conjecture 4] Fix the ensemble: for a scale X , let $\mathbb{E}_X$ average over odd $n \in (X, 2X]$ of the given residue class mod 8 with weight proportional to $1/n$ (the logarithmic sampling measure, as at Conjectures 5 and 4). Define, per class, the clean-minus-contaminated difference with explicit sign: $D(n) = R_o - R_e$ for $n \equiv 1 \pmod{8}$ (even lane contaminated), $D(n) = R_e - R_o$ for $n \equiv 3 \pmod{8}$, and $D(n) = R_e - R_o$ (sign immaterial) on the null classes $n \equiv 5, 7 \pmod{8}$. Set $D_{\text{sys}}(n) = [\sum_{(q,k): q^2+2k^2=n, q \text{ prime}, k \geq 1} \log q] / \lambda_{\text{an}}(n)$ —the weight is $\Lambda(q^2) = \log q$, each representation counted once in the $k \geq 1$ convention (the ordered-representation factor 2 of Conjecture 4 has no counterpart here, and doubling would be a

normalization error)—where $\lambda_{\text{an}}(n)$ is the analytic lane mean of $\log p$ over the candidate profile, defined with no reference to which candidates are prime:

$$\lambda_{\text{an}}(n) = \frac{\int_0^{\sqrt{(n-2)/2}} du}{\int_0^{\sqrt{(n-2)/2}} \frac{du}{\log(n-2u^2)}},$$

the mean of $\log(n-2u^2)$ under the density proportional to $1/\log(n-2u^2)$ on the candidate interval (the upper limit keeps $n-2u^2 \geq 2$, the convention of $I(N)$, avoiding the nonintegrable point $n-2u^2 = 1$), which is the profile a Bateman–Horn heuristic assigns to the prime representations. The definition is out-of-sample: the verification’s empirical lane mean over the observed prime representations is an estimator of λ_{an} , used as a consistency check, not part of the definition. The law, coefficient-identifying as at Conjecture 4(i): on the contaminated classes

$$\mathbb{E}_X[D - D_{\text{sys}}] = o(\mathbb{E}_X[D_{\text{sys}}]) \quad (X \rightarrow \infty),$$

and on the null classes—where D_{sys} vanishes identically, so that a comparison against it would be empty—the statement is made against the positive comparison scale

$$A(X) := \mathbb{E}_X^{(1)}[D_{\text{sys}}],$$

the logarithmic-ensemble mean of the systematic census taken over the contaminated class $n \equiv 1 \pmod{8}$: writing D_{null} for the difference D on a null class $n \equiv 5$ or $7 \pmod{8}$, the claim is $\mathbb{E}_X[D_{\text{null}}] = o(A(X))$.

Conjecture 13 (Microscopic variance law; classical family, the constant is Montgomery–Soundararajan’s [19]). Fix $\lambda > 0$ and $c \in (0, 1]$, and let X count primes in $[t, t + \lambda \log x)$ for t uniform on $[x, (1+c)x]$. Then, under the Hardy–Littlewood k -tuple conjectures, $X \Rightarrow \text{Poisson}(\lambda)$ (Gallagher’s argument [18], which is conditional on exactly those conjectures), and—conditional on the quantitative hypothesis spelled out after the statement (uniform Hardy–Littlewood with $o(h)$ control of the averaged singular series), without which the display below is a registered extrapolation rather than a consequence—at finite x

$$\frac{\text{Var } X}{\mathbb{E} X} = 1 - \frac{\log(\lambda \log x) + \gamma + \log 2\pi - 1}{\log x} + o\left(\frac{1}{\log x}\right),$$

with $\gamma + \log 2\pi - 1 = 1.41509\dots$, the leading correction being independent of the sampling constant c . [Interpolation clause] More generally, for windows of length $H = \lambda(\log x)^\alpha$ the same formula with $\log H$ in place of $\log(\lambda \log x)$,

$$\frac{\text{Var } X}{\mathbb{E} X} = 1 - \frac{\log H + \gamma + \log 2\pi - 1}{\log x} + o\left(\frac{1}{\log x}\right),$$

holds uniformly for $1 \leq \alpha \leq A$ (any fixed A with $H \leq x^{o(1)}$): one expression interpolates from Gallagher’s Poisson boundary ($\alpha = 1$, deficit $\asymp \log \log x / \log x$) through the mesoscopic Montgomery–Soundararajan regime (deficit $\asymp \alpha \log \log x / \log x$), with no transition other than the slow growth of $\log H$.

Conjecture 14 (The null-mechanism race; apparently new). *For the quadratic twin pairs of Conjecture 10 at $d = 2$, square contamination is algebraically impossible: $n^2 + 1 = q^2$ has no solution with $n \geq 1$, and $n^2 + 3 = q^2$ only at $(n, q) = (1, 2)$. The surviving classes of $n \bmod 5$ are $\{0, 1, 4\}$ (the single escape $n = 2$, the pair $(5, 7)$ in the killed class 2, is one bounded term), and the classes 1 and 4 have identical singular series (elementary, via CRT). Let $\pi_a^q(x)$ count solutions with $n \equiv a \pmod{5}$, $D(x) = \pi_1^q(x) - \pi_4^q(x)$, and define the normalized race process $Y(x) = D(x)/\sqrt{\pi_1^q(x) + \pi_4^q(x)}$. Then, in contrast to the twin race of Conjecture 1, this race is driftless: (i) $\mathcal{M}_x(D(t) \log^2 t / \sqrt{t}) \rightarrow 0$ at the drift-scale normalization of Conjecture 1(i)—the scale at which the contaminated races converge to nonzero constants—so this is the genuine negation of Conjecture 1(i) for this pair (the weaker normalization $\mathcal{M}_x(Y) \rightarrow 0$ with $Y = D/\sqrt{\pi^q}$ would not exclude a drift at the contamination scale, hence would not make this race a control; the form stated excludes exactly that); (ii) [occupation law, conjectured directly, with a separately falsifiable local hypothesis] the process is defined precisely at event index: let $n_1 < n_2 < \dots$ enumerate the n with $n^2 + 1$ and $n^2 + 3$ both prime and $n \equiv 1$ or $4 \pmod{5}$, set $\xi_i = +1$ if $n_i \equiv 1$ and -1 if $n_i \equiv 4$, and let $S_N = \xi_1 + \dots + \xi_N$ (so $D(x) = S_{N(x)}$ with $N(x)$ the number of such events to x). The sequence (ξ_i) is deterministic, so the hypothesis is stated for a randomly sited window rather than for S_N itself. Fix a growth scale N_0 and a window length $N = N(N_0)$ with $N = o(N_0)$, and choose the starting index M of the window according to the logarithmic measure on $[1, N_0]$:*

$$\Pr[M = m] = \left(\sum_{1 \leq m' \leq N_0} \frac{1}{m'} \right)^{-1} \frac{1}{m}, \quad 1 \leq m \leq N_0.$$

The hypothesis is an almost-sure invariance principle (ASIP) for the induced random path $(S_{M+j} - S_M)_{0 \leq j \leq N}$ under that sampling law: on a probability space carrying M and a standard Brownian motion B ,

$$S_{M+j} - S_M = B(j) + O(N^{1/2-\delta}) \quad \text{uniformly for } 0 \leq j \leq N,$$

almost surely in M as $N_0 \rightarrow \infty$, for some $\delta > 0$. The logarithmic sampling law is the same clock in which the occupation statement below is read, and randomizing the window start is what supplies the deterministic step sequence with a probability space. The weak functional limit theorem— $S_{\lfloor N\tau \rfloor} / \sqrt{N}$ converging weakly to standard Brownian motion on $\tau \in [0, 1]$ —is retained as the weaker, separately falsifiable consequence of the windowed ASIP. The occupation law itself, however, is not a consequence of the windowed hypothesis, and is conjectured directly. The windowed path $S_{M+j} - S_M$ is recentred at the window start, and recentring erases exactly the absolute level that occupation measures: a deterministic bias $b(n) = \sqrt{n}/\log n$ added to S_n shifts the level by $\sqrt{n}/\log n$ while contributing window increments of size $\sim j/(2\sqrt{n} \log n) \ll \sqrt{j}$, invisible to the windowed ASIP at every window scale, yet it would drive the occupation fraction to 1. A global coupling $D(t) = B(N(t)) + o(\sqrt{N(t)})$ with $N(t) \asymp t/\log^2 t$ would suffice, but for the deterministic sequence (ξ_i) no randomization is available that makes such a coupling well posed, so we do not hypothesize it. The Brownian computation is instead the motivation that identifies the limit and the constant: for a coupled walk, in logarithmic time $u = \log t$ the Lamperti reduction $Z(s) = e^{-s/2} B(e^s)$ is a stationary, ergodic Ornstein–Uhlenbeck process, and the logarithmic occupation measure of leadership converges almost surely to $\frac{1}{2}$. The conjectured law is that conclusion itself:

$$\frac{1}{\log x} \int_2^x \mathbf{1}_{\{D(t) > 0\}} \frac{dt}{t} \rightarrow \frac{1}{2},$$

with fluctuations of the Gaussian scale $\sqrt{\log 2 / \log x}$ (the occupation-indicator covariance $\frac{1}{2\pi} \arcsin e^{-u/2}$ integrates to $\frac{1}{2} \log 2$; note that the integrated covariance for the sign average $21 - 1$ is four times this, its standard deviation twice, the occupation fraction itself carrying $\sqrt{\log 2 / \log x}$); the classical arcsine spread survives only in the natural event clock, i.e. for occupation fractions of $\{B(m) > 0, m \leq N(x)\}$. A persistent systematic leader—a log-occupation stuck near 0 or 1 beyond the stated Gaussian scale—is the refutation.

Conjecture 15 (The cubic shift family: the distribution of its constants; instance-level law apparently unstated). For non-cube $a \geq 1$ let $C(a) = C(x^3 + a)$. Only $p \equiv 1 \pmod{3}$ moves the product, and the local root count there takes three values as a varies mod p : $\omega = 3$ with probability $(p-1)/3p$ ($-a$ a nonzero cube), $\omega = 1$ with probability $1/p$ (the case $p \mid a$, easy to overlook and essential to the mean-value identity below), and $\omega = 0$ with probability $2(p-1)/3p$; for $p \equiv 2 \pmod{3}$ the cube map is a bijection and $\omega = 1$ identically. Then: (i) [mean-value one, elementary lemma] $\mathbb{E}_a[\omega(p)] = 3 \cdot \frac{p-1}{3p} + \frac{1}{p} = 1$ exactly at every p , so the derived mean of $C(a)$ is exactly 1—the cubic analogue of the Korevaar–te Riele mean-value-one theorem for linear prime-pair constants (arXiv:0806.1667), here a one-line computation at each finite level; passing the mean through the infinite product needs more than almost-sure convergence, and the needed ingredient is on hand: the convergent variance sum makes the partial products an L^2 -bounded martingale (each factor has mean 1 and is independent of the earlier ones), so they are uniformly integrable and the limit law has mean exactly 1, not merely limit-of-means 1; (ii) [limit law] the empirical law of $\{C(a) : a \leq A, a \text{ a non-cube}\}$ converges, as $A \rightarrow \infty$, to the law of the random Euler product $\prod_{p \equiv 1(3)} f_p(\xi_p)$ with the three-state local variables ξ_p above—independent at any finite set of primes exactly, by CRT, with the infinite product’s tail controlled by the convergent variance sum (unlike the quadratic family of Conjecture 10(iii), no normalization is needed)—with derived standard deviation $0.2762\dots$ computed from the full three-state law; (iii) [uniformity] $\#\{n \leq N : n^3 + a \text{ prime}\} = C(a)I_a(N)(1 + o(1))$ uniformly over non-cube $a \leq (\log N)^B$.

Conjecture 16 (Conjecture F as a family; the core is previously stated, [1], cf. [23]). For odd A let $Q_A(N) = \#\{n \leq N : n^2 + n + A \text{ prime}\}$ and $C(A) = C(x^2 + x + A)$. Then, for every fixed $B > 0$, $Q_A(N) = C(A)I_A(N)(1 + o(1))$ uniformly over odd $1 \leq A \leq (\log N)^B$, with the residual field obeying the analogue of Conjecture 8(ii), kernel derived and probability space specified: for t uniform on $[N, 2N]$ and window length $H \leq N^{o(1)}$, let $Q_A(t; H) = \#\{t < n \leq t + H : n^2 + n + A \text{ prime}\}$ (mean $\asymp H/2 \log N$, which sets the regime). Two members share the index window, so with $C(A, A') = C(x^2 + x + A, x^2 + x + A')$,

$$\text{Cov}_t(Q_A, Q_{A'}) \sim [C(A, A') - C(A)C(A')] \frac{H}{4 \log^2 N},$$

with $C(A, A') = 0$ (hence negative correlation) for the locally exclusive pairs. This is the same-index contribution only: the full covariance adds the off-index pinned sum

$$\frac{1}{4 \log^2 N} \sum_{0 < |h| \leq H} (H - |h|) [C(f_A(x), f_{A'}(x + h)) - C(A)C(A')],$$

the two-parameter analogue of Conjecture 2’s $G(H)$, whose evaluation is this family’s open component. (The order of this sum matters: the joint event here is two single prime values at distinct indices, of probability $\asymp 1/4 \log^2 N$, unlike Conjecture 8’s pair-of-pairs events at $1/\log^4 x$. At that order the off-index sum, which grows like a pinned G -average over $\asymp H$ shifts against the same-index term’s single H , is generically the larger component, so the numerical

cancellation of the same-index kernel establishes nothing about the full cross-member covariance, and the localization of the observed profile deficit to the diagonal is an open question rather than an inference.) For $H/\log N \rightarrow \lambda$ the counts are jointly Poisson with local exclusions, and for $H/\log N \rightarrow \infty$ each fixed finite collection of studentized counts is asymptotically jointly normal with independent limiting coordinates, all correlations of size $\asymp 1/\log N$ making both kernels finite- N corrections to independence. As at Conjecture 8(ii), both distributional conclusions are asserted under an explicit hypothesis beyond the displayed covariances—uniform Hardy–Littlewood tuple estimates over the family, strong enough that every joint cumulant of order at least three of the studentized counts vanishes in the respective normalizations—since the covariance kernels alone do not determine the limit laws. (Uniformity over a range growing with N is the substantive claim; uniformity over any fixed finite set of A is logically automatic from the individual asymptotics.) Among odd $A \leq 199$, Euler’s $A = 41$ has the largest constant, $C(41) = 6.6395463\dots$ (Cohen’s high-precision value [31]).

Conjecture 17 (Twin-member Goldbach, orientation-resolved; the basis conjecture and the integral kernel are Dubner’s [9]; the orientation decomposition apparently new). *For even n let $R_T(n)$ count ordered (a, b) , $a + b = n$, with a and b both members of twin pairs, each ordered role (lower/upper) counted. Each of the four role assignments is a 4-form linear system in $t = a$ with root set $\{0, \pm 2\} \cup \{n, n \pm 2\}$ -type mod each p , hence its own n -dependent singular series $\mathfrak{S}_4^{(o)}(n)$, and*

$$R_T(n) = \left[\sum_o \mathfrak{S}_4^{(o)}(n) \right] \int_5^{n-5} \frac{dt}{\log^2 t \log^2(n-t)} (1 + o(1)),$$

the four constants computed exactly per n from the coincidence pattern of the root sets (which of $n, n \pm 2, n \pm 4$ each small prime divides). The error term is stated as $(1+o(1))$, not $(1+O(1/\log n))$: our own measurements show a second-order deficit whose $1/\log n$ coefficient is not yet pinned down (below). Two precision notes: the asymptotic runs over varying n , so what is assumed is a Hardy–Littlewood estimate for 4-form systems uniform in the n -dependent coefficients—strictly stronger than the fixed-system conjecture—and the number 5, the unique prime that is simultaneously a lower twin member (of $(5, 7)$) and an upper one (of $(3, 5)$), is double-counted consistently by the ordered-role convention, affecting finitely many representations per n and no asymptotic.

4 Instances and structural companions

Conjecture 18 (Contamination in prime triplets; apparently new). *The contamination calculus of Conjecture 1(v) extends to k -tuples, and the triplet pattern $(n, n + 2, n + 6)$ is its first instance beyond pairs. Of the three configurations placing a prime square inside the pattern, two die algebraically for $q > 3$ — $(q^2, q^2 + 2, q^2 + 6)$ by $3 \mid q^2 + 2$, and $(q^2 - 6, q^2 - 4, q^2)$ because $q^2 - 4 = (q - 2)(q + 2)$ is composite (at $q = 3$ the factorization is trivial, $q^2 - 4 = 5$ is prime, and the single bounded exceptional configuration $(3, 5, 9)$ occurs—one term, harmless to every asymptotic)—leaving the single doubly-thinned survivor $(q^2 - 2, q^2, q^2 + 4)$, which requires $q^2 - 2$ and $q^2 + 4$ simultaneously prime and forces $q \equiv \pm 2 \pmod{5}$ for $q \neq 5$ (else $5 \mid q^2 + 4$; at $q = 5$ all three conditions do hold, the configuration being $(23, 25, 29)$, but its start $23 \equiv 3 \pmod{5}$ lies outside the triplet start classes $\{1, 2\}$, so the single term is harmless). Triplet starts lie in classes $n \equiv 1, 2 \pmod{5}$ (single bounded exception $(5, 7, 11)$, start $\equiv 0$), and the surviving configuration has start $q^2 - 2 \equiv 2$: class 2 is contaminated and class 1 leads, in the drift-scale form*

of Conjecture 1(i) (a normalization at the noise scale, $\mathcal{M}_x((D - T_3)/\sqrt{\pi_3})$, would be vacuous for the coefficient, exactly as at the pair races):

$$\mathcal{M}_x\left(\frac{(\pi_3(t; 5, 1) - \pi_3(t; 5, 2)) \log^3 t}{\sqrt{t}}\right) \rightarrow c_3 := \lim_{x \rightarrow \infty} \frac{T_3(x) \log^3 x}{\sqrt{x}}, \quad T_3(x) = \frac{1}{\log^3 x} \sum_{\substack{q \leq \sqrt{x}, q \neq 5 \\ q^2-2, q^2+4 \text{ prime}}} \log(q^2-2) \log q \log$$

(c_3 exists and is the Bateman–Horn constant of the triple $(q, q^2 - 2, q^2 + 4)$, since the census sum is $\sim c_3 \sqrt{x}$; convergence of the logarithmic mean at this normalization is the same Rubinstein–Sarnak-type structure hypothesis as at Conjecture 1(i)), with drift $\asymp \sqrt{x}/\log^3 x$, one logarithm weaker than the pair races, and drift-to-noise $\asymp \log^{-3/2} x$ against their $\log^{-1}$, a scale the calculus itself predicts and the verification must respect. The mechanism clause and the averaging clause are asserted separately, as at Conjecture 1(i).

Conjecture 19 (The contamination matrix: sexy pairs with two surviving orientations; apparently new). For the pattern $(n, n+6)$, both prime-square orientations survive, on complementary classes of $q \bmod 5$ —the first matrix instance of the calculus (every pair race so far had exactly one): orientation $A = (q^2 - 6, q^2)$ requires $q^2 - 6$ prime, forcing $q \equiv \pm 2 \pmod{5}$ for $q \neq 5$ (at $q = 5$ the value 19 is prime, but its start $19 \equiv 4 \pmod{5}$ lies outside the sexy start classes $\{1, 2, 3\}$), and lands in start class 3 (mod 5), class 3 (mod 8); orientation $B = (q^2, q^2 + 6)$ requires $q^2 + 6$ prime, forcing $q \equiv \pm 1 \pmod{5}$ (the term $q = 5$ falls outside the start classes), and lands in class 1 (mod 5), class 1 (mod 8). Sexy starts occupy classes $\{1, 2, 3\} \bmod 5$ and $\{1, 3, 5, 7\} \bmod 8$; with $T_A(x) = \frac{1}{\log^2 x} \sum_{q^2-6 \text{ prime}, q \neq 5} \log q \log(q^2 - 6)$, $T_B(x) = \frac{1}{\log^2 x} \sum_{q^2+6 \text{ prime}, q \neq 5} \log q \log(q^2 + 6)$, and c_A, c_B their drift-scale limits ($c_A = \lim T_A \log^2 x / \sqrt{x}$, the Bateman–Horn constant of $(q, q^2 - 6)$, likewise c_B for $(q, q^2 + 6)$), the predicted drift vector, in the drift-scale limiting-log-mean sense of Conjecture 1(i) (an $\mathcal{M}_x$ -over- $\sqrt{\pi}$ normalization would be coefficient-blind), is: mod 5: $\mathcal{M}_x((\pi(2) - \pi(3)) \log^2 t / \sqrt{t}) \rightarrow c_A$, $\mathcal{M}_x((\pi(2) - \pi(1)) \log^2 t / \sqrt{t}) \rightarrow c_B$ (class 2 clean); mod 8: $\mathcal{M}_x((\frac{1}{2}(\pi(5) + \pi(7)) - \pi(3)) \log^2 t / \sqrt{t}) \rightarrow c_A$, $\mathcal{M}_x((\frac{1}{2}(\pi(5) + \pi(7)) - \pi(1)) \log^2 t / \sqrt{t}) \rightarrow c_B$ (classes 5, 7 clean, mutually symmetric, with $\mathcal{M}_x((\pi(5) - \pi(7)) \log^2 t / \sqrt{t}) \rightarrow 0$). Two independently computable drift constants, one certified null class per modulus. The mechanism clause and the averaging clause are asserted separately, as at Conjecture 1(i).

Conjecture 20 (Fibonacci–Lucas twins: the convergent side stress-tested; the object is OEIS A080327). (i) [rank-disjointness, elementary, with its exception stated] every odd prime factor r of L_p has rank of apparition exactly $2p$ (from $z(r) \mid 2p$, $z(r) \nmid p$, and $z(r) = 2$ being impossible), and every prime factor of F_p ($p \neq 5$ prime) has rank exactly p : the odd divisor pools of the two numbers are disjoint. The prime 2 is the unique exception: 2 divides both F_p and L_p precisely when $p = 3$ ($F_3 = 2$, $L_3 = 4$; $\gcd(F_p, L_p) = 2$ iff $3 \mid p$)—the restriction to odd divisors is essential, the clause being false at $p = 3$ without it. Disjoint pools remove shared-prime correlation but not all correlation: dependence can still enter through the order structure of the recurrence, which is why (iii) matters; (ii) [finiteness] only finitely many primes p have F_p and L_p simultaneously prime; (iii) [calibration clause, quantitative] the naive joint accounting—probability $(e^\gamma \log p / (p \log \phi))^2$ per index—assigns the catalogued index $p = 148091$ (OEIS A080327; both F_p and L_p are probable primes—numbers of roughly 30,950 digits, with Lucas numbers proved prime only up to index 56003—so everything recorded here is conditional on both probable-prime classifications, for which no Baillie–PSW pseudoprime is known: no unconditional refutation is claimed from uncertified numbers) prior mass about 1.4×10^{-2} beyond 10^4 . That rare-event datum is recorded descriptively, with no significance claim attached to it: the model was examined after the event was known,

and no test protocol was fixed in advance. A single surprising event is not a strict refutation, and it does not by itself estimate the size of the correction: an underestimated constant, genuine cross-dependence, and a mis-calibrated finite-range tail are competing explanations that one event cannot separate. What the datum does bear on (conditionally as above) is not the model but the list: it is fatal to any Goldilocks completeness list; the conjectural content of (ii) is that even the corrected accounting has convergent sum.

Conjecture 21 (Factorial twins; the uniqueness core is previously stated, OEIS A088054; clauses (i) and (iii) are ours). (i) [window rigidity, elementary] for $n \geq 4$ and $2 \leq |a| \leq n$ the number $n! + a$ is composite (any prime factor p of a satisfies $p \leq n$, so $p \mid n! + a$, and $n! + a > n \geq p$); the two small anomalies $2! - 2 = 0$ and $3! - 3 = 3$ are the only exceptions, which is why the clause is restricted to $n \geq 4$. Hence among all offsets of bounded size the only candidates for primality near $n!$ are $a = \pm 1$: the twin question below is the unique bounded-offset constellation question at $n!$, and every admissible offset set is a subset of $\{-1, +1\}$. (ii) [uniqueness; previously stated] Only finitely many n have $n! - 1$ and $n! + 1$ simultaneously prime, and the complete list is $n = 3$: i.e. $6 = 3! = 3\#$ is the unique factorial lying between twin primes. (iii) [joint fluctuation model] the two single-sided counts $F_{\pm}(N) = \#\{n \leq N : n! \pm 1 \text{ prime}\}$ are each $\sim e^{\gamma} \log N$ and, under the independent-indices model stated below, asymptotically independent, with $F_+ - F_-$ normalized by $\sqrt{2e^{\gamma} \log N}$ asymptotically standard normal. The probability space is declared: the clause is about a random model in which the events $E_n^{\pm} = \{n! \pm 1 \text{ prime}\}$ carry the Caldwell–Gallot marginal hazards e^{γ}/n with an unspecified dependence structure, and the Gaussian limit is asserted under three explicit hypotheses on that structure: the same-index bound $\sum_{n \leq N} |\text{Cov}(\mathbf{1}_{E_n^+}, \mathbf{1}_{E_n^-})| = o(\log N)$, which pins the variance at $2e^{\gamma} \log N (1 + o(1))$ and which the joint law $\sim (e^{\gamma}/n)^2$ derived below supplies with a convergent sum—without it the remaining hypotheses leave the second cumulant free, and a model with $\Pr(E_n^+ \cap E_n^-) = e^{\gamma}/2n$ halves the limiting variance; the cross-index covariance bound $\sum_{m < n \leq N} |\text{Cov}(\mathbf{1}_{E_m}, \mathbf{1}_{E_n})| = o(\log N)$ over the four sign pairs (the variance itself is $\asymp \log N$), and—since pairwise control alone cannot preclude higher-order dependence—the higher-cumulant condition $\kappa_r(F_+ - F_-) = o((\log N)^{r/2})$ for each fixed $r \geq 3$, the method-of-moments sufficient condition. Neither is imported from an independence model; whether they hold for the arithmetic dependence across factorial indices is the open component of this clause.

Theorem 1 (Cube obstruction). For $k \geq 2$, the cube k^3 is representable as $p + j^3$ with p prime, $j \geq 1$, if and only if $3k^2 - 3k + 1$ is prime. Consequently the set of integers not representable as $p + k^3$ is infinite: the non-representable cubes alone have counting function $\sim x^{1/3}$. (That the full exceptional set is $\sim x^{1/3}$ follows only when combined with Conjecture 22(i), and is conjectural.)

Proof. $k^3 - j^3 = (k - j)(k^2 + kj + j^2)$; for $1 \leq j \leq k - 2$ both factors exceed 1, so the difference is composite. The only candidate is $j = k - 1$, giving $3k^2 - 3k + 1$. Since $3k^2 - 3k + 1$ is composite for a density-one set of k (all but $O(K/\log K)$ of $k \leq K$, by an upper-bound sieve of Brun or Selberg type applied to the quadratic’s prime values), all but a vanishing proportion of cubes are unrepresentable. $\square$

Conjecture 22 (The boundary trichotomy for polynomial ladders; the divisibility principle is classical and its cubic case is Cunningham’s cuban observation (OEIS A002407); the family classification is apparently new). For $F \in \mathbb{Z}[x]$ with $\deg F \geq 2$ and positive leading coefficient, and $m > j \geq 1$, $(m - j) \mid F(m) - F(j)$ (textbook), and the divided difference $(F(m) - F(j))/(m - j)$ —whose leading form is then positive on the cone $m > j \geq 1$ —exceeds 1 outside an effectively

bounded region; so $F(m) - F(j)$ prime forces $m - j = 1$ apart from finitely many exceptional pairs (for the family below the exceptional set is provably empty). Both hypotheses are necessary: for $\deg F = 1$ the cofactor can be identically 1 and the collapse fails, and for negative leading coefficient the divided difference tends to $-\infty$ on the cone, so the “exceeds 1” clause would be false as stated (one may equivalently keep general sign and read the claim through $|F(m) - F(j)|$). Every representation problem $F(m) = p + F(j)$ in this range thus collapses to the boundary polynomial $D_F(m) = F(m) - F(m - 1)$. The content is the classification of the boundary lanes. For $F = x^3 + cx$, $c \geq 0$: (i) [trichotomy, elementary] $D_F = 3m^2 - 3m + 1 + c$ and the lane is dead-parity for c odd (D_F always even), dead-3-adic for even $c \equiv 2 \pmod{3}$ ($3 \mid D_F$ always), and admissible exactly for $c \equiv 0, 4 \pmod{6}$; (ii) [uniform boundary law] over the admissible lanes, $\#\{m \leq M : D_F(m) \text{ prime}\} \sim C(D_F) I(M)$ uniformly for $c \leq (\log M)^B$ —a Bateman–Horn family whose members are indexed by the ladder classification; (iii) [attributed core] the case $c = 0$ is the cuban ladder: Cunningham’s classical observation that a prime difference of cubes forces consecutive arguments, with the non-cube exceptional set of $n = p + k^3$ finite (Hardy–Littlewood’s $E_3(X) = O(1)$ [1]).

Conjecture 23 (The power-obstruction ladder: an elementary structural proposition with a Bateman–Horn corollary; apparently new as a family). For $k \geq 2$ and $m \geq 2$, write $D_k(m) = m^k - (m - 1)^k$, and call m^k representable if $m^k = p + j^k$ for some prime p and $j \geq 1$. Then: (i) [theorem] for composite k , no k -th power is representable: if r is a proper divisor of k with $r > 1$ (such r exists precisely because k is composite; the excluded value $r = 1$ would allow only the useless factor $m - j = 1$ at $j = m - 1$), then for every $1 \leq j < m$ the difference $m^k - j^k$ has the proper factor $m^r - j^r > 1$ with cofactor > 1 ; (ii) [elementary] for prime k (including $k = 2$), m^k is representable if and only if $D_k(m)$ is prime; and D_k is irreducible for prime k (a root α has $(\alpha/(\alpha - 1))^k = 1$ with $\alpha/(\alpha - 1) \neq 1$, so $\zeta = \alpha/(\alpha - 1)$ is a primitive k -th root of unity and $\mathbb{Q}(\alpha) = \mathbb{Q}(\zeta)$ has degree $k - 1 = \deg D_k$, forcing irreducibility; explicitly $D_k(x) = (x - 1)^{k-1} \Phi_k(x/(x - 1))$. The forcing $j = m - 1$ needs only $k \geq 2$); (iii) [conjecture] for each prime $k \geq 3$ the representable lane follows Bateman–Horn ($k = 2$ is a theorem, the prime number theorem on the lane $2m - 1$): $\#\{m \leq M : D_k(m) \text{ prime}\} \sim C_k \int_2^M dt / \log D_k(t)$).

Conjecture 24 (The alternating cyclotomic chain; apparently new). There are infinitely many primes p such that, writing $u = \Phi_3(p) = p^2 + p + 1$, the three integers

$$p, \quad u = p^2 + p + 1, \quad \Phi_6(u) = u^2 - u + 1 = p^4 + 2p^3 + 2p^2 + p + 1$$

are all prime, with $\#\{p \leq x\} \sim C_6^{\text{ch}} I(x)$ for the Bateman–Horn system $\{x, x^2 + x + 1, x^4 + 2x^3 + 2x^2 + x + 1\}$, $C_6^{\text{ch}} = 3.6143 \pm 0.011$. The first chains are $(2, 7, 43)$ and $(3, 13, 157)$.

Conjecture 25 (Twin cyclotomic bases: the complete quadratic family; no prior trace found). For the three cyclotomic polynomials of degree 2 (indices k with $\varphi(k) = 2$, i.e. $k \in \{3, 4, 6\}$), consider $\Phi_k(n)$ and $\Phi_k(n + 1)$ simultaneously prime. Then: (i) [parity obstruction, elementary] for $k = 4$ one member of the pair $(n^2 + 1, n^2 + 2n + 2)$ is even for every n , so the only prime pair is $n = 1$, giving $(2, 5)$ —the same obstruction that forbids consecutive $n^2 + 1$ primes; (ii) [collapse, elementary] $\Phi_6(x) = \Phi_3(x - 1)$ identically, so the $k = 6$ pair at index n is the $k = 3$ pair at index $n - 1$: the family contains exactly one nontrivial statement; (iii) [conjecture, the live instance] there are infinitely many n for which the length-3 repunits in bases n and $n + 1$ are simultaneously prime: $\#\{n \leq N : \Phi_3(n) = n^2 + n + 1 \text{ and } \Phi_3(n + 1) = n^2 + 3n + 3 \text{ both prime}\} \sim C_5 I(N)$, with $C_5 = C(x^2 + x + 1, x^2 + 3x + 3)$.

Part II

Structural conjectures in five programmes

5 Programme I: connected prime-pattern fields

Let $H \subset \mathbb{Z}$ be finite and admissible, and write

$$\Lambda_H(n) = \prod_{h \in H} \Lambda(n+h), \quad Y_H(x, L) = \sum_n w\left(\frac{n-x}{L}\right) (\Lambda_H(n) - \mathfrak{S}(H)),$$

where w is a fixed smooth compactly supported function and $\mathfrak{S}(H)$ is the Hardy–Littlewood singular series $\prod_p (1 - \nu_H(p)/p)(1 - 1/p)^{-|H|}$, with $\nu_H(p)$ the number of residue classes occupied by H modulo p . The window position x is sampled from $[X, 2X]$ with density $dx/(x \log 2)$ and $X^\varepsilon \leq L \leq X^{1-\varepsilon}$. For translated motifs $H_i + t_i$, their *exact overlap type* records every equality among shifted prime constraints. The connected singular series $\mathfrak{S}^c(K_1, \dots, K_r)$ is obtained by Möbius inversion over set partitions of the indexed tuple $(K_1, \dots, K_r)$.

Conjecture 26 (Connected motif generating functional). *Fix admissible motifs $H_1, \dots, H_m$ and put*

$$\mathcal{Z}_{X,L}(\mathbf{z}) = \mathbb{E}_x \exp\left(\sum_{i=1}^m z_i Y_{H_i}(x, L)\right).$$

For every fixed total degree R , the degree- $\leq R$ Taylor polynomial of $\log \mathcal{Z}_{X,L}$ has a uniform asymptotic expansion indexed by connected exact-overlap diagrams of at most R translated motifs. A diagram whose coincidence hypergraph connects the r translated sets $H_{i_1} + t_1, \dots, H_{i_r} + t_r$ contributes, at the leading order of its stratum,

$$W_\tau(L; \mathbf{t}) \mathfrak{S}\left(\bigcup_{j=1}^r (H_{i_j} + t_j)\right) \prod_v (\log X)^{m_v-1},$$

where $W_\tau(L; \mathbf{t}) = L \int_{\mathbb{R}} \prod_{j=1}^r w(u + t_j/L) du$ is the archimedean window-overlap weight of the diagram and m_v is the multiplicity of the prime constraint at v : any set partition that separates a coincidence loses at least one factor of $\log X$, so the Möbius alternation leaves the single-block moment as the leading term of a connected diagram. The same alternation cancels the leading terms of disconnected diagrams, whose contribution is carried by the connected singular series of their disjoint blocks at strictly smaller logarithmic order, reducing for the fully disjoint type to $\mathfrak{S}^c(H_{i_1} + t_1, \dots, H_{i_r} + t_r)$ with no logarithmic enhancement. The log powers are therefore partition dependent and are never factored across the alternation, and every prime-power counterterm is included in the local weight attached to its multiplicity. Ordering terms lexicographically by powers of L and $\log X$, the remainder after any fixed truncation is smaller than the last retained scale, uniformly in the mesoscopic range.

Conjecture 27 (Complete overlap-renormalization filtration). *Fix a finite motif family $H_1, \dots, H_m$. Apply the degree-two specialization of Conjecture 26 and order all of its connected covariance strata by their asymptotic scales $s_1 \succ s_2 \succ \dots$ (lexicographically in powers of L , $\log X$, and the renormalized disjoint scales). In the mesoscopic range scales sharing the same total power of L and of logarithms differ only by powers of $\theta = \log L / \log X$ and merge into a single stratum whose limiting form is polynomial in θ . The filtration, the graded convergence, and the eigenvalue*

asymptotics below are asserted for every fixed $\theta \in (\varepsilon, 1 - \varepsilon)$, with each $A_\nu = A_\nu(\theta)$ nondegenerate for every such θ . Recursively subtract the earlier strata and let A_ν be the limiting covariance form at scale s_ν , and put

$$\mathcal{W}_\nu = \bigcap_{\mu < \nu} \ker A_\mu.$$

For every fixed truncation of this ordered diagram expansion, the spectral projections of the covariance matrix converge to the associated graded spaces $\mathcal{W}_\nu / \mathcal{W}_{\nu+1}$, and every eigenvalue whose first nonzero term occurs within the truncation is asymptotic to s_ν times a positive eigenvalue of the induced form A_ν . Vectors annihilating all retained forms pass canonically to the next connected diagram scales of Conjecture 26.

Consequently every asymptotic variance scale of a finite motif statistic is generated by a connected overlap or renormalized disjoint diagram from Conjecture 26, and no scale external to that diagrammatic expansion occurs. The filtration is invariant under replacing the motif family by one with isomorphic exact-overlap incidence algebra and identical local Euler weights.

Conjecture 28 (Regularized mesoscopic local-spectral trace formula). *Fix motifs H_i, H_j , a Schwartz function ϕ with $\phi(0) = 0$, an even cutoff $\eta \in C_c^\infty(\mathbb{R})$ with $\eta = 1$ near the origin, and an even mollifier $\kappa \in C_c^\infty(\mathbb{R})$ of integral one. Let $Q_{ij}^{\text{loc}}(\phi; L)$ be the exact-overlap-renormalized singular-series form*

$$\sum_{t \in \mathbb{Z}} \left[L \int_{\mathbb{R}} w(u) w(u + t/L) du \right] \phi(t/L) \mathfrak{S}^c(H_i, H_j + t).$$

Let $\Lambda_{T,\eta}^\sharp$ be the Weil explicit-formula distribution in the logarithmic variable, with each nontrivial zero of ordinate γ multiplied by $\eta(\gamma/T)$ and with the pole, trivial-zero, archimedean, and prime-power terms kept as separately named summands. For $\delta > 0$, put

$$\Lambda_{T,\eta,\delta}^\sharp = \Lambda_{T,\eta}^\sharp *_{\log} \kappa_\delta, \quad \kappa_\delta(u) = \delta^{-1} \kappa(u/\delta).$$

This is a smooth function, so products of its shifted copies are unambiguous. Define

$$Q_{ij}^{\text{spec}}[\eta, \kappa] = Q_{ij}^{\text{spec}}(\phi; X, L; T, \delta, \eta, \kappa)$$

by inserting these mollified copies into the smoothed motif covariance, applying the same exact-overlap Möbius projection as on the local side, and subtracting the explicitly specified pole, diagonal, trivial-zero, archimedean, and prime-power counterterms before the mollifier is removed.

For every sufficiently large fixed A, B , take

$$T = (X/L)(\log X)^A, \quad \delta = (L/X)(\log X)^{-B}.$$

Then

$$Q_{ij}^{\text{spec}}(\phi; X, L; T, \delta, \eta, \kappa) - Q_{ij}^{\text{loc}}(\phi; L) = o(V_{ij}(X, L))$$

uniformly for $X^\varepsilon \leq L \leq X^{1-\varepsilon}$ and for ϕ in bounded Schwartz sets, where V_{ij} is the first nonzero disjoint covariance scale. The same limit holds jointly for every fixed matrix of motifs and tests. Changing A, B, η , or κ within the stated admissible class changes the renormalized spectral form by $o(V_{ij})$, so the limiting trace functional is canonical.

Conjecture 29 (Anchored arithmetic polymer expansion for prime motifs). *Fix an admissible motif H of size k . A finite set $A \subset \mathbb{Z}$ of occurrence starts is a polymer when $0 \in A$, the overlap graph on A is connected, and $U_A = \bigcup_{t \in A} (H + t)$ is admissible. Define its anchored codimension*

$$d_H(A) = |U_A| - |H|;$$

this is the number of new prime constraints beyond the occurrence anchored at 0. In the regularized spectral model of Conjecture 28, let Z be the limiting low-mode vector (the joint limit, asserted as part of this conjecture, of the mode integrals of the Conjecture 28 functional against frequencies below a fixed cutoff K on the L -scale, the Conjecture 28 limits taken first and $K \rightarrow \infty$ last), let $\lambda(t \mid Z)$ be the conditional one-point occurrence intensity, and set

$$\lambda_H^0 = \frac{\mathfrak{S}(H)}{(\log X)^k}, \quad \tilde{\lambda}(t \mid Z) = \lambda(t \mid Z) / \lambda_H^0.$$

For every fixed polymer A , uniformly for bounded diameter,

$$\frac{(\log X)^{d_H(A)} \rho^{(|A|)}(A \mid Z)}{\lambda(0 \mid Z) \prod_{t \in A \setminus \{0\}} \tilde{\lambda}(t \mid Z)} \longrightarrow \frac{\mathfrak{S}(U_A)}{\mathfrak{S}(H)}$$

in probability in the successive limits defining Z . Thus the conditional environment supplies only the one-point intensity, while every connected Palm activity is graded by $(\log X)^{-d_H(A)}$ and weighted by the complete-union singular series.

If admissible overlapping translates exist, put

$$\delta(H) = \min_{\substack{t \neq 0 \\ H \cup (H+t) \text{ admissible}}} (|H \cup (H+t)| - |H|), \quad \mathcal{T}_H = \{t \neq 0 : H \cup (H+t) \text{ admissible}, |H \cup (H+t)| - |H| = \delta(H)\}$$

Then the cluster-start fraction has the first nontrivial expansion

$$\theta_X(H) = 1 - \frac{c_H}{(\log X)^{\delta(H)}} + o((\log X)^{-\delta(H)}), \quad c_H = \sum_{\substack{t \in \mathcal{T}_H \\ t < 0}} \frac{\mathfrak{S}(H \cup (H+t))}{\mathfrak{S}(H)}.$$

The set $\mathcal{T}_H$ is symmetric under $t \mapsto -t$, and the cluster-start fraction counts, for each occurrence, only whether it has an overlapping predecessor, so the sum is restricted to $t < 0$ and equals half the symmetric sum over $\mathcal{T}_H$. If no admissible overlap exists, $\theta_X(H) = 1 + o((\log X)^{-M})$ for every fixed M . Higher corrections are the connected Ursell sums of anchored polymers ordered by $d_H(A)$. These finite- X Gibbs expansions are projectively consistent and subcritical, and after division by the low-zero intensity the cluster-start process is Poisson to each fixed order of this triangular expansion.

Conjecture 30 (Topological expansion of non-Gaussian prime cumulants). *For every connected exact-overlap diagram in Conjecture 26, expand its connected singular series over the linking primes and form the linkage graph of each term, whose vertices are the motif occurrences and whose edges join two occurrences sharing a linking prime in that term. Let g be the first Betti number of the linkage graph. Once all larger coincidence strata have been subtracted, the total contribution of the terms of Betti number g is smaller than that of the spanning-tree terms with the same external motifs by at least $(\log L)^{-g}$. The complete leading coefficient at each logarithmic order is the sum of the linked-term weights with the corresponding Betti number. In particular,*

the first nonzero odd cumulant of a balanced motif statistic, a combination Y_c whose coefficient vector annihilates every coincidence-stratum form of Conjecture 27, is carried by incidence trees. For the single-prime field the fully spread connected part $\kappa_{2m+1}^{\text{sp}}$, defined by subtracting every coincidence stratum as in Conjecture 27, obeys

$$\kappa_{2m+1}^{\text{sp}} = (c_m + o(1))L^m(\log L)^{m+1},$$

where c_m is the total connected tree weight. The raw cumulant adds the explicit coincidence diagrams of Conjecture 26, which carry powers of $\log X$ and are computable term by term, so the tree law is a statement about the spread diagrams, exactly the regime of Kuperberg's singular-series sums.

6 Programme II: arithmetic first-arrival fields and class groups

For a prime modulus q and $a \in \mathbb{F}_q^\times$, let $p(q, a)$ be the least prime in the class a and put

$$T_q(a) = \frac{\text{Li}(p(q, a))}{q-1}, \quad \mathcal{N}_{q,a} = \sum_{p \equiv a(q)} \delta_{\text{Li}(p)/(q-1)}.$$

The probability space chooses q uniformly among primes in $[Q, 2Q]$ and then chooses a uniformly from $\mathbb{F}_q^\times$. Write $e(x) = e^{2\pi i x}$.

Conjecture 31 (Connected first-arrival functional). *Let $w_{r,Q}^c$ be the connected factorial-cumulant measure of the random point process $\mathcal{N}_{q,a}$. For every fixed $r \geq 2$ there is a signed measure K_r , locally finite on the off-diagonal region $\{u \in \mathbb{R}_+^r : u_i \text{ distinct}\}$, such that, on that region (the diagonal strata carry strictly smaller normalizations, as in the stratification of Conjecture 26, and are excluded here),*

$$(\log Q)^{r-1} w_{r,Q}^c \implies K_r$$

vaguely, and K_1 denotes the limiting intensity measure, which in this clock is Lebesgue measure on $\mathbb{R}_+$. The measure K_r is obtained by averaging, over the modulus aspect, the connected Hardy–Littlewood densities of r primes constrained to one residue class. For every nonnegative compactly supported f in a fixed uniform ball on which the Laplace functional is analytic,

$$\log \mathbb{E} \exp(-\langle f, \mathcal{N}_{q,a} \rangle) = \sum_{r \geq 1} \frac{(-1)^r}{r!} \int \prod_{j=1}^r (1 - e^{-f(u_j)}) dw_{r,Q}^c(\mathbf{u}),$$

and the asymptotic expansion obtained by inserting the limits K_r is uniform to every fixed connected order. In particular the entire survival function, not merely its first correction, is determined by the hierarchy $(K_r)_{r \geq 2}$.

Conjecture 32 (Tested Gauss-polyspectral reciprocity for the least-prime field). *Set $T_q(0) = \text{Li}(q)/(q-1)$ and centre the resulting field f_q on all of $\mathbb{F}_q$. Let $\mathbf{A}_q$ and $\mathbf{M}_q$ be its complete additive and multiplicative Fourier coefficient vectors, including the explicit rank-one coordinate needed to record the value at 0. There is then an explicit invertible Gauss matrix U_q with*

$$\mathbf{M}_q = U_q \mathbf{A}_q.$$

Choose q uniformly among primes in $[Q, 2Q]$. For every finite family of degrees $r_1, \dots, r_m$ and uniformly bounded fibrewise test tensors $\Phi_{j,q}$, define the scalar additive observables

$$X_{j,Q}(q) = \langle \Phi_{j,q}, \mathbf{A}_q^{\otimes r_j} \rangle$$

and the transported multiplicative observables

$$Y_{j,Q}(q) = \left\langle (U_q^{-1})^{*\otimes r_j} \Phi_{j,q}, \mathbf{M}_q^{\otimes r_j} \right\rangle.$$

For every modulus, $X_{j,Q}(q) = Y_{j,Q}(q)$, so all joint cumulants under the random-modulus law satisfy the exact finite identity

$$\text{Cum}_Q(X_{1,Q}, \dots, X_{m,Q}) = \text{Cum}_Q(Y_{1,Q}, \dots, Y_{m,Q}).$$

This is the probability-space-correct form of Gauss transport. It makes no reference to a cumulant of a deterministic coefficient vector.

The conjectural content is that, after normalization by the first nonzero connected scale, tests supported on additive zero-sum tensors converge to contractions of the Fourier transforms of the arrival kernels K_r from Conjecture 31. Transporting those tests fibrewise by the exact Gauss matrices and only then taking the modulus average gives the same limits as the corresponding multiple explicit-formula correlations of Dirichlet L -function zeros. The two descriptions determine the same continuous multilinear functional on the completion of the tested tensor class.

Conjecture 33 (Rubinstein–Sarnak terminal chaos dichotomy). Assume GRH and a q -aspect linear-independence/random-phase hypothesis for the low zeros of primitive Dirichlet L -functions. Let x_q satisfy $\text{Li}(x_q)/(q-1) = \log(q-1)$ and let $G_{q,A}(a)$ be the centred, variance-normalized low-zero field obtained from characters $\chi \neq \chi_0$ and zeros $|\gamma| \leq (\log q)^A$, with the standard compensator $-\frac{1}{2} \text{Var } G_{q,A}(a)$. Define the random empirical chaos measure

$$\mathcal{M}_{q,A}(dg) = \frac{1}{q-1} \sum_{a \in \mathbb{F}_q^\times} \exp\left(G_{q,A}(a) - \frac{1}{2} \text{Var } G_{q,A}(a)\right) \delta_{G_{q,A}(a)}(dg).$$

Let b_Q be the smallest level with $\mathbb{E}_q \#\{a : T_q(a) > b_Q\} \leq 1$. In the successive limits $Q \rightarrow \infty$ and then $A \rightarrow \infty$, the pair consisting of $\mathcal{M}_{q,A}$ and

$$\sum_a \delta_{(G_{q,A}(a), T_q(a) - b_Q)}$$

converges jointly. Conditionally on the limiting chaos measure $\mathcal{M}$, the terminal cluster-start process is Poisson with directing measure

$$e^{-x} \mathcal{M}(dx).$$

At finite modulus the cluster-start thinning enters through the terminal extremal index $\theta_{\text{occ}}(q)$, the analogue for the last classes of the cluster-start fraction $\theta_X(H)$ of Conjecture 29, together with the terminal connected-polymer decorations of Conjecture 31. Both corrections enter at the first nonzero connected scale and vanish in the successive limit, so $\theta_{\text{occ}}(q) \rightarrow 1$ and the limiting directing measure carries no cluster factor.

The nature of the directing measure is governed by the covariance-energy statistic

$$\mathfrak{E}_{q,A} = \frac{1}{(q-1)^2} \sum_{a,b \in \mathbb{F}_q^\times} \text{Cov}(G_{q,A}(a), G_{q,A}(b))^2.$$

If $\mathfrak{E}_{q,A} \rightarrow 0$ along the successive limit, $\mathcal{M}$ self-averages to the deterministic exponentially tilted Gaussian law and the terminal process is an ordinary decorated Poisson process. If $\liminf \mathfrak{E}_{q,A} > 0$ and the Gaussian-chaos second moments are uniformly integrable, $\mathcal{M}$ is nondegenerate and the limit is a genuine Cox process. No nondegeneracy is asserted without this criterion.

Conjecture 34 (Nonlinear spectral response calculus for arithmetic first arrivals). *Let a sequence of modulus ensembles have a small explicit-formula perturbation of the arrival intensity*

$$V_Q(a, t) = \sum_{\chi \in \mathcal{C}_Q} v_{Q,\chi}(t) \chi(a), \quad \|V_Q\|_T \rightarrow 0,$$

where $\mathcal{C}_Q$ has uniformly bounded cardinality and, on every fixed interval $[0, T]$, the canonical norm is $\|V\|_T = \sup_{a, 0 \leq t \leq T} |V(a, t)|$. Assume that the connected hierarchy of Conjecture 31 satisfies an exponential cluster bound there, so its Laplace functional is analytic in a fixed $\|\cdot\|_T$ -ball. If $S_Q(a, t)$ and $S_Q^{(0)}(t)$ are the perturbed and ordinary survival functions, then for every fixed R

$$\log \frac{S_Q(a, t)}{S_Q^{(0)}(t)} = \sum_{m=1}^R \mathcal{R}_m[V_Q^{\otimes m}](a, t) + o(\|V_Q\|_T^R)$$

uniformly on compact t -ranges. The causal Volterra operator $\mathcal{R}_m$ is the universal linked-cluster contraction of the full hierarchy $(K_r)_{r \geq 1}$ with m marked perturbation insertions, and truncating the connected hierarchy at order J gives a convergent approximation as $J \rightarrow \infty$.

If the input character support is $\mathcal{C}_Q$, the m th response is supported exactly on products of at most m input characters (subject only to coincidences and the trivial character). In particular the linear response preserves character rank, while the first new harmonics occur at quadratic order with coefficients determined by connected three-point arrival correlations.

Conjecture 35 (Local-information capacity of odd class groups). *Fix an odd prime ℓ . Sample negative fundamental discriminants $D \in [-2Y, -Y]$, conditioned on a fixed coarse two-primary invariant. For a set of odd primes S_Y , put $M_Y = \prod_{p \in S_Y} p$, let $\Sigma_Y(D) = ((D/p))_{p \in S_Y}$ be the attainable sign vector, and let H_Y be the logarithm of the number of attainable cells. Let $\delta_Y(\sigma)$ be the exact CRT density of the sign cell after imposing quadratic reciprocity and the fixed two-primary data (for a single odd unramified prime the density of each sign is $p/(2(p+1))$, and an unconditioned cell density is the corresponding product), and let $\mathcal{D}_Y$ be the ambient discriminant family. Write*

$$\Delta_Y(S_Y) = \sup_{\sigma} \left| \frac{\#\{D \in \mathcal{D}_Y : \Sigma_Y(D) = \sigma\}}{|\mathcal{D}_Y| \delta_Y(\sigma)} - 1 \right|$$

over attainable cells with $|\mathcal{D}_Y| \delta_Y(\sigma) \geq \log Y$.

There is a sequence $\varepsilon_Y \rightarrow 0$, independent of S_Y in the subcritical range, such that uniformly over those cells

$$d_{\text{TV}}(\mathcal{L}(\text{Cl}(D)[\ell^\infty] \mid \Sigma_Y = \sigma, 2\text{-data}), \mathcal{L}_{\text{CL}, \ell}) \leq \varepsilon_Y + O(\Delta_Y(S_Y)),$$

and the analogous bound holds for every fixed Cohen–Lenstra moment. In particular, for every $\varepsilon > 0$, the conclusion is uniform whenever

$$H_Y \leq (1 - \varepsilon) \log Y, \quad \log M_Y \leq \left(\frac{1}{2} - \varepsilon\right) \log Y, \quad \Delta_Y(S_Y) = o(1).$$

Since every odd prime contributes at most $\log 3 \leq \log p$ of cell entropy, $H_Y \leq \log M_Y$ throughout, so the conductor constraint is the binding one in the displayed range, while the entropy governs the impossibility clause below. At $H_Y \geq (1 + \varepsilon) \log Y$ no uniform statement is possible. Up to the conductor scale at which Δ_Y becomes macroscopic, every first-order failure of local independence factors through this discriminant-cell discrepancy, and there is no earlier odd-class-group obstruction.

7 Programme III: finite logarithms, regulators, and algebraic tori

For $p \nmid a$, write

$$q_p(a) = \frac{a^{p-1} - 1}{p} \pmod{p}.$$

For a generating tuple $g = (g_1, \dots, g_s)$, let $R(g)$ be its multiplicative relation lattice and let $\mathbb{T}_g \subset (\mathbb{R}/\mathbb{Z})^s$ be the identity component of its annihilator. Its dimension is the saturated multiplicative rank.

Conjecture 36 (Kummer–Haar law on relation tori). *Let $\Gamma \leq \mathbb{Q}^\times$ be finitely generated and torsion-free, saturated modulo torsion (if $x^n \in \Gamma \cdot \{\pm 1\}$ for some $n \geq 1$ then $x \in \Gamma \cdot \{\pm 1\}$), and let g be any ordered generating tuple. As p varies in every fixed compatible cyclotomic–Kummer Chebotarev class, the vectors*

$$\frac{(q_p(g_1), \dots, q_p(g_s))}{p} \in (\mathbb{R}/\mathbb{Z})^s$$

equidistribute with Haar measure on $\mathbb{T}_g$. More strongly, for every finite collection of compatible Chebotarev restrictions, the conditional empirical measures converge to the Haar disintegration on the corresponding components, and the support is contained in no proper closed coset of $\mathbb{T}_g$. The law is invariant under changing generators and depends on Γ only through its saturation and the fixed Kummer conditioning data.

Conjecture 37 (Rank-dimensional large sieve for finite logarithms). *Choose a Smith-normal-form basis of the character lattice $X^*(\mathbb{T}_g) = \mathbb{Z}^s / R(g)^{\text{sat}}$. There is a constant $\delta_\Gamma > 0$ such that, for every $\varepsilon > 0$, every $M \leq X^{\delta_\Gamma}$, and all complex weights a_p supported on $X < p \leq 2X$,*

$$\sum_{\substack{m \in X^*(\mathbb{T}_g) \\ \|m\|_\infty \leq M}} \left| \sum_{X < p \leq 2X} a_p e\left(\frac{\langle m, q_p(g) \rangle}{p}\right) \right|^2 \ll_{\Gamma, \varepsilon} (M^{\text{rank } \Gamma} + \pi(X)) X^\varepsilon \sum_p |a_p|^2.$$

The same inequality holds after every fixed compatible cyclotomic–Kummer restriction, and for matrix-valued weights with the Hilbert–Schmidt norm. The exponent δ_Γ is invariant under changing the generating tuple.

Conjecture 38 (Mesoscopic-to-lattice shrinking-target transition). *Let $\Gamma \leq \mathbb{Q}^\times$ be torsion-free of rank r and saturated modulo torsion, and choose a basis of Γ , identifying its relation torus with $(\mathbb{R}/\mathbb{Z})^r$. For integers w_p with $1 \leq w_p = o(p)$, let $E_p(w)$ be the event that every coordinate of the Fermat-quotient vector has a representative of absolute value at most w_p .*

Mesoscopic universality. *If $w_p \rightarrow \infty$, then in every range in which the Haar mass*

$$\lambda(X) = \sum_{p \leq X} \left(\frac{2w_p + 1}{p} \right)^r$$

tends to infinity, the count of $E_p(w)$ is asymptotic to $\lambda(X)$. On prime blocks with bounded Haar mass, the randomly translated event process converges to a Poisson process with that mass.

Lattice transition. *For every fixed finite target set $B \subset \mathbb{Z}^r$, there is an arithmetic regulator intensity $\kappa_{\Gamma, B} \geq 0$ such that*

$$\#\{p \leq X : q_p|_\Gamma \in B\} = (\kappa_{\Gamma, B} + o(1)) \sum_{p \leq X} p^{-r}$$

whenever the sum diverges, while only finitely many such primes occur when $r \geq 2$. As a separately falsifiable genericity clause, for Kummer-generic Γ (the Kummer fields $\mathbb{Q}(\zeta_m, \Gamma^{1/m})$ have the maximal degree $\varphi(m)m^r$ for every m) and $B = \{0\}$ one has $\kappa_{\Gamma, B} = 1$. The transition from Haar universality to the arithmetic intensity occurs only when w_p remains bounded.

For a p -adic unit a , write $\langle a \rangle_p$ for the Teichmüller lift of its residue, and for a prime $\mathfrak{p} \mid p$ of a number field write $\tau_{\mathfrak{p}}$ for the Teichmüller lift in the completion at $\mathfrak{p}$. Define $q_{p,k}(a) = p^{-1} \log(a \langle a \rangle_p^{-1}) \pmod{p^k}$. The matrix entries below are the $\mathfrak{p}$ -adic analogues of $q_{p,k}$; on $\mathbb{Q}$ with $k = 1$ one has $q_{p,1}(a) \equiv -q_p(a) \pmod{p}$, the Fermat quotient up to sign.

Conjecture 39 (Global-lattice horizontal p -adic regulator-matrix law). *Let $K/\mathbb{Q}$ be a fixed Galois number field with group G , and let $\Gamma \leq \mathcal{O}_K^\times$ be torsion-free and saturated modulo $\mu(K)$, with ordered basis $\epsilon_1, \dots, \epsilon_s$. Define the fixed global relation lattice*

$$\mathcal{R}_\Gamma = \left\{ (c_{\sigma j}) \in \mathbb{Z}^{G \times s} : \prod_{\sigma \in G} \prod_{j=1}^s \sigma(\epsilon_j)^{c_{\sigma j}} \in \mu(K) \right\}.$$

Let $\mathbb{T}_\Gamma \subset (\mathbb{R}/\mathbb{Z})^{G \times s}$ be its annihilator. This lattice contains the norm and Galois relations and is independent of the chosen basis up to the natural integral change of coordinates.

For a rational prime p splitting completely in K , label the primes above p by G , use the Teichmüller lift in each completion, and form the depth- k matrix

$$L_{p,k}(\Gamma) = \left(p^{-1} \log_{\mathfrak{p}}(\epsilon_j \tau_{\mathfrak{p}}(\bar{\epsilon}_j)^{-1}) \pmod{p^k} \right)_{\mathfrak{p} \mid p, 1 \leq j \leq s}.$$

It lies in the annihilator of $\mathcal{R}_\Gamma \otimes \mathbb{Z}/p^k \mathbb{Z}$. As p varies through any fixed compatible Frobenius class, $L_{p,1}/p$ equidistributes with Haar measure on $\mathbb{T}_\Gamma$, and the depth- k matrices are Haar at every fixed depth k , with the depth- $(k+1)$ law pushing forward to the depth- k law under reduction (the moduli vary with p , so this is a per-depth statement with compatible marginals, not Haar on one fixed group).

Let $\mathcal{D}$ be a basis-invariant determinantal or Smith stratum in this projective module, and let $\mu_{p,k}(\mathcal{D})$ be its exact local Haar mass. The primes for which $L_{p,k} \in \mathcal{D}$ have counting law governed

by $\sum_{p \leq X} \mu_{p,k}(\mathcal{D})$. For a determinantal or Smith stratum of codimension c , whose point counts are polynomial in p , $\mu_{p,1}(\mathcal{D}) = p^{-c} + O(p^{-c-1})$, and codimension one has a Poisson point process in $\log \log p$ after uniform random translation, while codimension at least two has a summable large-prime tail together with the exact small-prime atoms.

Conjecture 40 (Functorial horizontal finite logarithms on algebraic tori). *Let $T/\mathbb{Q}$ be an algebraic torus, let $\mathcal{T}$ be its smooth integral model away from a finite set, and let $\Gamma \leq T(\mathbb{Q})$ be finitely generated. Let S be the identity component of the Zariski closure of Γ . At a good unramified prime p , let $\tau_p(\bar{P})$ be the canonical prime-to- p torsion lift of the reduction of $P \in \Gamma$ and define*

$$\ell_p(P) = p^{-1} \log_T(P \tau_p(\bar{P})^{-1}) \pmod{p} \in \text{Lie}(S)(\mathbb{F}_p).$$

Here $\text{Lie}(S)(\mathbb{F}_p)$ is identified with $(\mathbb{Z}/p)^{\dim S}$ through a fixed integral basis of the Lie algebra of $\mathcal{T}$, with representatives in $[0, p)$; a different basis changes the matrix by a fixed element of $\text{GL}_{\dim S}(\mathbb{Z})$, which preserves the Haar law and the shrinking-target intensities. For any generators $P_1, \dots, P_s$ of Γ , the normalized matrix $(\ell_p(P_i)/p)_i$ equidistributes, in every compatible Frobenius class, with Haar measure on the real relation subtorus cut out jointly by the algebraic subgroup S and the integral relations among the P_i . For every morphism of tori $\phi : T \rightarrow T'$,

$$\ell_p(\phi(P)) = d\phi(\ell_p(P)),$$

and the horizontal Haar measures push forward under the induced map. Isogenies preserve the shrinking-target intensities after the explicit finite kernel correction.

8 Programme IV: arboreal arithmetic dynamics

For a degree- d map f and basepoint a , write $K_n = \mathbb{Q}(f^{-n}(a))$, $G_n = \text{Gal}(K_n/\mathbb{Q})$, and let $\mu_n(C) = |C|/|G_n|$ on conjugacy classes. For primes $p \leq X$ unramified in K_n , let $\hat{\mu}_{n,X}$ be the empirical Frobenius measure.

Conjecture 41 (Entropy–conductor profile of arboreal resolution). *For a degree- d map f and level n , let $G_{n,f}$ be the arboreal quotient and $\mu_{n,f}$ its Haar measure on conjugacy classes. For N independent samples from $\mu_{n,f}$ define the exact sampling obstruction*

$$\mathcal{S}_{n,f}(N) = \mathbb{E} d_{\text{TV}} \left(N^{-1} \sum_{j=1}^N \delta_{Z_j}, \mu_{n,f} \right).$$

For each nontrivial irreducible Artin representation ρ of $G_{n,f}$, let $q(\rho)$ be its Artin conductor and define the GRH conductor energy

$$\mathcal{A}_{n,f}(X) = \frac{\log X}{\sqrt{X}} \left(\sum_{\rho \neq 1} (\dim \rho)^2 (\log q(\rho) + \dim(\rho) \log X)^2 \right)^{1/2}.$$

This is a canonical character-theoretic replacement for a single discriminant proxy. For a parameter family $\mathcal{F}(B)$ put

$$n_{\mathcal{F}}^*(B, X; \eta) = \max \left\{ n : \mathbb{E}_{f \in \mathcal{F}(B)} [\mathcal{S}_{n,f}(\pi(X)) + \mathcal{A}_{n,f}(X)] \leq \eta \right\}.$$

Assume the generic arboreal image is open, exceptional parameters have density zero, Artin holomorphy and GRH hold at the levels considered, and the family is in the large-sieve range of Conjecture 42. Then the actual empirical resolution depth, defined by the same total-variation threshold with prime Frobenius samples, differs from $n_{\mathcal{F}}^*(B, X; \eta)$ by $O_{\mathcal{F}, \eta}(1)$.

More precisely, suppose, uniformly for a density-one set of parameters, that the class measures have a limiting complete Rényi profile

$$\psi_{\mathcal{F}}(\alpha) = \lim_{n \rightarrow \infty} d^{-n} \log \sum_C \mu_{n,f}(C)^\alpha$$

uniformly for α in a fixed interval containing $[1/2, 2]$, and the corresponding Artin-conductor energies have finite positive exponential profiles on the same d^n scale. Then

$$n_{\mathcal{F}}^*(B, X; \eta) = \log_d \log X + O_{\mathcal{F}, \eta}(1).$$

Two families of the same degree with boundedly different complete Rényi and conductor profiles have resolution depths differing by $O(1)$, and in particular a fixed-index open subgroup changes only the bounded shift.

Conjecture 42 (Arboreal family large sieve and cumulant independence). *Let $\mathcal{F} \rightarrow \mathbb{A}^1$ be a generically finite-index arboreal family. For each level n and irreducible representation ρ of the generic quotient G_n , let $\mathcal{V}_{n,\rho}$ be the associated middle-extension sheaf on the good parameter locus. Define the geometric conductor budgets*

$$\mathfrak{C}_{n,r} = \sum_{\rho_1, \dots, \rho_r} \left(\prod_{j=1}^r \dim \rho_j \right)^2 \operatorname{cond} \left(\boxtimes_{j=1}^r \mathcal{V}_{n,\rho_j} \right)^2, \quad \mathfrak{C}_n = \mathfrak{C}_{n,1},$$

where the external tensor product is taken on the CRT parameter space for distinct primes, and the conductor includes rank, singularities, tame drop, and Swan terms.

Square-root tier. For every $\varepsilon > 0$ and all complex coefficients $\alpha_{p,\rho}$,

$$\sum_{|c| \leq B} \left| \sum_{p \leq X} \sum_{\rho \neq 1} \alpha_{p,\rho} \operatorname{tr} \rho(\operatorname{Frob}_{p,c}) \right|^2 \ll_{\varepsilon, \mathcal{F}} (B + X^2 \mathfrak{C}_n^{1+\varepsilon}) \sum_{p,\rho} |\alpha_{p,\rho}|^2,$$

uniformly when $X^2 \mathfrak{C}_n^{1+\varepsilon} \leq B^{1-\varepsilon}$, after deleting the singular fibres.

Connected tier. For every fixed $r \geq 3$, centred trace functions at distinct primes have family-averaged connected cumulants with square-root cancellation in the parameter-family count, uniformly in the range $X^r \mathfrak{C}_{n,r}^{1+\varepsilon} \leq B^{1-\varepsilon}$, and equivalently, after normalization, every fixed connected Frobenius cumulant tends to zero. An exponential connected-moment bound makes this convergence uniform in r over the range needed for occupancy functionals.

Consequently, whenever the analytic Chebotarev error is $o(1)$ but the iid sampling functional of μ_n is nonvanishing, the complete conjugacy-cylinder count vector, averaged over parameters, has the nonuniform multinomial occupancy law of independent samples from μ_n , including its missing-mass, collision, and total-variation profile. The multinomial conclusion is derived from the connected tier, not from the L^2 bound alone.

Conjecture 43 (Coloured dynatomic Galois-factor process). *For $f_c(x) = x^d + c$, let the critical exact-period polynomial factor as $\Psi_n(c) = \prod_{j=1}^{s_n} F_{n,j}(c)$ over $\mathbb{Z}$. Sample c from a dyadic height interval away from postcritically finite, discriminant, and fixed-resultant loci.*

For every fixed n , condition on any fixed finite local valuation state and normalize the prime factors of each $F_{n,j}(c)$ by that component's residual logarithmic mass. The component-coloured factor processes converge jointly to independent Frobenius-marked Poisson–Dirichlet PD(1) processes, with the mark in colour j given by the fixed-point-size-biased conjugacy law in the splitting field of $F_{n,j}$. The total logarithmic mass of ramified factors tends to zero.

Uniformly for $n = n(H)$ with $\deg \Psi_n = o(\log H)$, every fixed correlation functional whose support lies inside the available level of distribution converges to the corresponding restricted-support correlation of these coloured marked processes. No full-process claim is made at growing degree without level-one distribution.

Conjecture 44 (Complexity-uniform primitive valuation process for dynatomic values). *For the critical exact-period polynomial $\Psi_n(c)$, remove its canonical greatest common divisor in $\mathbb{Q}[c]$ with the earlier critical-orbit product and denote the result by Ψ_n^{new} . Define the logarithmic arithmetic complexity*

$$\mathfrak{H}_n = \deg \Psi_n^{\text{new}} + \log^+ H_{\text{coeff}}(\Psi_n^{\text{new}}) + \log^+ |\text{Disc}(\Psi_n^{\text{new}})| + \sum_{m < n} \log^+ |\text{Res}(\Psi_n^{\text{new}}, f_c^m(0))|,$$

where zero resultants have already been removed by the canonical gcd. For each prime p , let $\nu_{n,p}$ be the exact Haar law on $\mathbb{Z}_p$ of the primitive valuation

$$V_{n,p}(c) = v_p(\Psi_n^{\text{new}}(c)) \mathbf{1}_{p \nmid f_c^m(0)} \quad \forall m < n.$$

All discriminant, resultant, and collision primes remain inside these local laws.

For every control function $\omega(H) = o(\log H)$, uniformly over levels with $\mathfrak{H}_n \leq \omega(H)$, the finite-prime valuation vectors for c in a dyadic height interval converge to the corresponding marginals of $\bigotimes_p \nu_{n,p}$. In addition, for every $\eta > 0$,

$$\lim_{z \rightarrow \infty} \limsup_{H \rightarrow \infty} \sup_{n: \mathfrak{H}_n \leq \omega(H)} \mathbb{P} \left(\sum_{p > z} (\log p) (V_{n,p} - 1)_+ > \eta \log |\Psi_n^{\text{new}}(c)| \right) = 0,$$

and the analogous unnormalized tail $\sum_{p > z} \mathbb{P}(V_{n,p} \geq 2)$ tends to zero whenever the local squarefree product is positive. These two clauses give convergence in the topology generated by bounded cylinder functions and the logarithmic squarefull-mass functional.

Consequently

$$\mathbb{P}(\Psi_n^{\text{new}}(c) \text{ is squarefree}) = \prod_p \tilde{\nu}_{n,p}(\{0, 1\}) + o(1)$$

uniformly in the stated complexity range, where $\tilde{\nu}_{n,p} = \nu_{n,p}$ except at the finitely many collision primes dividing one of the resultants $\text{Res}(\Psi_n^{\text{new}}, f_c^m(0))$, at which $\tilde{\nu}_{n,p}$ is the law of the raw valuation $v_p(\Psi_n^{\text{new}}(c))$, and the logarithmic squarefull mass converges to the law determined by the complete local product.

Conjecture 45 (Divisor-sensitive classification of dynamical gcd height). *Let $F = (f, g)$ be a split map on $(\mathbb{P}^1)^2$, with f, g disintegrated (in the sense of Medvedev and Scanlon [34], so neither is linearly conjugate to a power map, a Chebyshev map, or a close relative of these) of the same degree, and let $P = (a, b)$ have positive canonical heights. Let D_1, D_2 be the two coordinate-zero divisors and let E be the exceptional divisor of the blow-up at $D_1 \cap D_2$. Then*

$$\limsup_{n \rightarrow \infty} \frac{h_E(F^n(P))}{\max\{h(f^n(a)), h(g^n(b))\}} > 0$$

if and only if there exist $m \geq 0$ and an irreducible F -periodic curve Z containing $F^m(P)$, with Z not contained in $D_1 \cup D_2$ (automatic when both coordinates have positive canonical height), such that, on the normalization of Z , the pullbacks of D_1 and D_2 have a common nonzero effective component. In the absence of such eventual entry into a divisor-compatible periodic curve, the normalized gcd height tends to zero.

The converse is unconditional in every setting where the required Vojta inequality for the blow-up is known, and otherwise is explicitly conditional on that inequality. The same criterion extends to split maps on $(\mathbb{P}^1)^r$ by replacing a common component with positive codimension-one intersection multiplicity among the pulled-back divisors.

9 Programme V: adelic factorization and sieve flow

Throughout this programme $P^+(m)$ and $P^-(m)$ denote the largest and smallest prime factors of m , the constant γ is Euler's, $\rho_f(p)$ counts the roots of f modulo p , and every sample $n \in [N, 2N]$ is drawn uniformly.

Conjecture 46 (Frobenius-marked Poisson–Dirichlet process). *Let $f \in \mathbb{Z}[x]$ be irreducible with splitting field L and transitive Galois group G . Sample $n \in [N, 2N]$. Mark each unramified prime factor p of $f(n)$ by its conjugacy class $C \subset G$ and give it mass $u = \log p / \log |f(n)|$. The macroscopic marked process converges to a marked PD(1) process with conditional mark law*

$$\mathbb{P}(C \mid u) = \frac{|C|}{|G|} \text{fix}(C), \quad \text{fix}(C) \text{ the number of fixed roots of an element of } C,$$

independent of u . Given the masses and every fixed finite local valuation state, the marks of finitely many macroscopic factors are asymptotically independent. Under a normal quotient equipped with the induced permutation representation, the marked process pushes forward functorially.

Conjecture 47 (Three-scale factorization of polynomial values). *Let $f \in \mathbb{Z}[x]$ be irreducible, sample $n \in [N, 2N]$, choose $y = y(N) \rightarrow \infty$ with $\log y = o(\log N)$, and write $R_y = f(n)/S_y(f(n))$, where $S_y(m) = \prod_{p \leq y} p^{v_p(m)}$ is the y -smooth part. Jointly with the complete small-prime valuation field $(v_p(f(n)))_{p \leq y}$, the residual logarithmic factor process*

$$\sum_{p \mid R_y} v_p(R_y) \delta_{(\log p / \log |R_y|, \text{Frob}_p)}$$

converges to the following conservation-corrected product object: the small valuations have their polynomial Kubilius law (independent exact local limit laws for the small-prime valuations), and conditional on their consumed logarithmic mass the residual process is a scale-invariant Poisson process of intensity du/u , marked by Conjecture 46, conditioned to have total mass one. The small field and the unconditioned residual Poisson process are asymptotically independent, and all leading dependence after conditioning is the single residual-mass constraint.

For disjoint normalized size bands bounded away from zero, the unconditioned residual process has independent increments. Its mesoscopic restriction and its ranked macroscopic atoms yield respectively the scale-invariant factor process and the Frobenius-marked PD(1) partition. Thus the local Kubilius field is the boundary variable fixing the available mass, while the mesoscopic and macroscopic laws are two projections of one residual process, and no additional cross-scale coupling survives.

Conjecture 48 (Adelic Gibbs gluing for reducible polynomial values). *Let $f = \prod_{i=1}^s g_i$ be squarefree with irreducible components, sample $n \in [N, 2N]$, and put $t = n/N$. For each prime p , let ν_p be the exact Haar law on $\mathbb{Z}_p$ of the valuation vector $(v_p(g_1(x)), \dots, v_p(g_s(x)))$. For $y \rightarrow \infty$ with $\log y = o(\log N)$, consider stable convergence jointly with the archimedean coordinate t and condition on $\mathcal{L}_y = (v_p(g_i(n)))_{p \leq y, i \leq s}$. After normalizing each component by its own residual logarithmic mass, the coloured Frobenius-marked factor processes converge to conditionally independent copies of the three-scale law of Conjecture 47, with the marks of Conjecture 46.*

Unconditionally, the joint process is the projective Gibbs mixture obtained by integrating those conditional product laws against Lebesgue measure in t and the adelic local law $\bigotimes_p \nu_p$, in the iterated order $N \rightarrow \infty$ and then $y \rightarrow \infty$. Every bounded finite-dimensional Laplace functional is determined by

$$Z_p(\mathbf{z}) = \int_{\mathbb{Z}_p} \prod_i z_i^{v_p(g_i(x))} dx$$

together with the archimedean size profile $(\log |g_i(Nt)|)_i$. No additional colour coupling survives after this adelic–archimedean mixture.

Conjecture 49 (Full multivariate adelic saddle law for joint smoothness). *Let $f_1, \dots, f_s \in \mathbb{Z}[x]$ be pairwise coprime and primitive, sample $n \in [N, 2N]$, and put $t = n/N$. For complex variables $\mathbf{s}$ and smoothness bounds $\mathbf{y}$, define*

$$\mathcal{Z}_{p,\mathbf{y}}(\mathbf{s}) = \int_{\mathbb{Z}_p} p^{-\sum_{i:p \leq y_i} (s_i-1)v_p(f_i(x))} dx, \quad \hat{\mathcal{Z}}_{p,\mathbf{y}}(\mathbf{s}) = \frac{\mathcal{Z}_{p,\mathbf{y}}(\mathbf{s})}{\mathcal{Z}_{p,\mathbf{y}}(+\infty)}.$$

The denominator is the exact Haar probability that p divides none of the relevant $f_i(x)$. Put

$$\Phi_t(\mathbf{s}) = \sum_p \log \hat{\mathcal{Z}}_{p,\mathbf{y}}(\mathbf{s}) + \sum_i (s_i - 1) \log |f_i(Nt)|.$$

For $f(x) = x$, this gives the usual truncated zeta factor $(1 - p^{-s})^{-1}$. Define the joint local probability by the Perron-calibrated multivariate inverse Mellin integral

$$\mathcal{P}_t^{\text{joint}} = \frac{1}{(2\pi i)^s} \int_{\boldsymbol{\sigma}_t + i\mathbb{R}^s} e^{\Phi_t(\mathbf{s})} \prod_{i=1}^s \frac{ds_i}{s_i},$$

with the standard truncated-contour limit, where $\boldsymbol{\sigma}_t$ is the unique real saddle in the admissible region. The factors s_i^{-1} are part of the canonical cumulative smoothness event, and for one variable they recover the Hildebrand–Tenenbaum Perron normalization [33].

Uniformly when every $u_i(t) = \log |f_i(Nt)| / \log y_i$ lies in a fixed compact saddle-point region,

$$\mathbb{P}(P^+(f_i(n)) \leq y_i \ \forall i) \sim \int_1^2 \mathcal{P}_t^{\text{joint}} dt.$$

The integral has the full multivariate saddle expansion. Writing $H_t = \nabla^2 \Phi_t(\boldsymbol{\sigma}_t)$, its leading term includes the joint action, the amplitude $\prod_i \sigma_{i,t}^{-1}$, and $(\det H_t)^{-1/2}$, with the usual lattice correction when the valuation span is proper.

If $\mathcal{P}_{i,t}^{\text{marg}}$ are the identically calibrated one-polynomial probabilities, then $\mathcal{P}_t^{\text{joint}} / \prod_i \mathcal{P}_{i,t}^{\text{marg}}$ is the complete ratio of joint and marginal actions, Perron amplitudes, Hessian determinants, and lattice corrections. Its Euler-product component is the ratio of the exact p -adic partition functions evaluated at the joint and marginal saddles. None of these factors may be replaced by a scalar Euler correction at the marginal saddle.

Conjecture 50 (Buchstab–Bateman–Horn component flow). *Let $f \in \mathbb{Z}[x]$ be primitive, irreducible, and admissible ($\rho_f(p) < p$ for every prime p), and sample $n \in [N, 2N]$. For y with $u_n = \log |f(n)| / \log y$ in a fixed compact subset of $(1, \infty)$, let $\Omega_y(f(n))$ be the number of prime factors of $f(n)$ exceeding y , counted with multiplicity, on the event $P^-(f(n)) > y$. Put*

$$V_f(y) = \prod_{p \leq y} \left(1 - \frac{\rho_f(p)}{p}\right).$$

Define the Buchstab component functions by

$$\omega_1(u) = \frac{1}{u}, \quad \omega_j(u) = \frac{1}{u(j-1)!} \int_{\substack{t_1, \dots, t_{j-1} \geq 1 \\ t_1 + \dots + t_{j-1} \leq u-1}} \frac{dt_1 \cdots dt_{j-1}}{t_1 \cdots t_{j-1}} \quad (j \geq 2),$$

so that $\sum_{j \geq 1} \omega_j(u) = \omega(u)$. Then for every fixed j ,

$$\mathbb{P}(P^-(f(n)) > y, \Omega_y(f(n)) = j) = V_f(y) e^\gamma \mathbb{E}_{n \in [N, 2N]} \omega_j(u_n) + o(V_f(y)),$$

uniformly on compact u -ranges. Conditional on the ordered logarithmic sizes of the j remaining factors, their unramified Frobenius marks are independent with the fixed-point-size-biased law of Conjecture 46. Summing over j gives the rough-value Buchstab law, $j = 1$ gives the Bateman–Horn prime density, and $j = 2$ gives the semiprime transition. Whenever $\max_{N \leq n \leq 2N} |f(n)| < y^2$, every rough value has exactly one prime factor above y , so the $j = 1$ component exhausts the rough set, while the density formula itself remains conjectural.

Part III

Six conjectures across fields

10 Relative polynomial pretentiousness

Let $P \in \mathbb{Z}[x]$ be primitive (the greatest common divisor of its coefficients is 1) and irreducible, with positive leading coefficient. For a prime p write

$$\rho_P(p) = \#\{a \in \mathbb{Z}/p\mathbb{Z} : P(a) \equiv 0 \pmod{p}\}.$$

The weight $\rho_P(p)/p$ is the first-order frequency with which p divides a polynomial value $P(n)$. A function f on the positive integers is completely multiplicative if $f(mn) = f(m)f(n)$ for all $m, n \geq 1$. For completely multiplicative f, g with values in the closed complex unit disc, define the truncated P -relative pretentious distance

$$\mathbb{D}_P(f, g; X)^2 = \sum_{p \leq X} \frac{\rho_P(p)}{p} (1 - \operatorname{Re}(f(p)\overline{g(p)})), \quad \mathbb{D}_P(f, g)^2 = \lim_{X \rightarrow \infty} \mathbb{D}_P(f, g; X)^2 \in [0, \infty].$$

Every summand is nonnegative, so the truncated distance is nondecreasing in X and the limit exists. Throughout, a primitive Dirichlet character χ of conductor c is regarded as the c -periodic completely multiplicative function on $\mathbb{Z}$ that vanishes on integers not coprime to c , so that expressions such as $\chi(P(a))$ are defined for every residue a modulo any multiple of c . For $t \in \mathbb{R}$,

the comparison function $n \mapsto \chi(n)n^{it}$ is completely multiplicative with values in the closed unit disc.

Let $P_1, \dots, P_m \in \mathbb{Z}[x]$ be pairwise distinct primitive irreducible polynomials with positive leading coefficients and no fixed prime divisor, meaning $\rho_{P_j}(p) < p$ for every j and every prime p . Assume multiplicative independence modulo constants:

$$\prod_{j=1}^m P_j(x)^{e_j} \in \mathbb{Q}^\times, \quad e_j \in \mathbb{Z} \implies e_1 = \dots = e_m = 0.$$

Write $\mathbb{S}^1$ for the unit circle in the complex plane.

Conjecture 51 (Relative polynomial pretentiousness inverse principle). *Let $f_1, \dots, f_m : \{1, 2, 3, \dots\} \rightarrow \mathbb{S}^1$ be completely multiplicative. If*

$$\limsup_{X \rightarrow \infty} \left| \frac{1}{\log X} \sum_{\substack{n \leq X \\ P_j(n) > 0 \ \forall j}} \frac{\prod_{j=1}^m f_j(P_j(n))}{n} \right| > 0,$$

then there exist primitive Dirichlet characters $\chi_1, \dots, \chi_m$ and real numbers $t_1, \dots, t_m$ such that the following three conditions hold simultaneously:

(i) $\mathbb{D}_{P_j}(f_j, \chi_j(n)n^{it_j}) < \infty$ for every $1 \leq j \leq m$,

(ii) $\sum_{j=1}^m t_j \deg P_j = 0$,

(iii) for some common multiple q of the conductors of $\chi_1, \dots, \chi_m$,

$$\frac{1}{q} \sum_{a \bmod q} \prod_{j=1}^m \chi_j(P_j(a)) \neq 0,$$

the characters being extended by 0 on nonunits as above.

11 Monotone threshold purification of vector paths

Let $d, m \geq 1$. Let $v_1, \dots, v_m \in \mathbb{R}^d$ satisfy $\|v_i\|_2 \leq 1$, where $\|\cdot\|_2$ is the Euclidean norm. Let $a_i : [0, 1] \rightarrow [0, 1]$ be continuous nondecreasing functions with $a_i(0) = 0$ and $a_i(1) = 1$, called a monotone activation schedule, and define the fractional path

$$F(t) = \sum_{i=1}^m a_i(t) v_i.$$

A threshold rounding of F is a choice of $\tau_1, \dots, \tau_m \in (0, 1]$ together with the pure path

$$G(t) = \sum_{i: \tau_i \leq t} v_i.$$

Conjecture 52 (Monotone threshold purification of vector paths). *There exists a universal constant $C < \infty$ such that for every $d, m \geq 1$, every $(v_i)_{i \leq m}$, and every monotone activation schedule $(a_i)_{i \leq m}$ as above, some threshold rounding satisfies*

$$\sup_{0 \leq t \leq 1} \|G(t) - F(t)\|_2 \leq C\sqrt{d}.$$

12 Entropy dimension of random-free graphons

A graphon is a symmetric measurable function $W : [0, 1]^2 \rightarrow [0, 1]$, and W is random-free if it takes values in $\{0, 1\}$ almost everywhere. Define the row pseudometric

$$d_W(x, y) = \int_0^1 |W(x, z) - W(y, z)| dz.$$

Identifying points at zero distance and completing yields a metric-measure space (Ω_W, d_W, ρ_W) , the row space, where ρ_W is the pushforward of Lebesgue measure. The row space is Ahlfors s -regular if it is compact and there exist $c, C, r_0 > 0$ such that

$$cr^s \leq \rho_W(B(x, r)) \leq Cr^s$$

for every $x \in \text{supp } \rho_W$ and every $0 < r < r_0$. Let $G(n, W)$ be the random labelled graph on $[n] = \{1, \dots, n\}$ obtained from independent uniform $U_1, \dots, U_n$ on $[0, 1]$ by declaring ij an edge exactly when $W(U_i, U_j) = 1$, and let $H(G(n, W))$ be its Shannon entropy with natural logarithms.

Conjecture 53 (Entropy dimension of random-free graphons).

- (i) [nonvacuity] *For every $s \in (0, 2]$ there exists a random-free graphon whose row space is Ahlfors s -regular.*
- (ii) [entropy-dimension law] *If the row space of a random-free graphon W is Ahlfors s -regular for some $s > 0$, then*

$$\lim_{n \rightarrow \infty} \frac{H(G(n, W))}{n \log n} = s.$$

Remark 1. *The refined assertion $H(G(n, W)) = sn \log n + O_W(n)$ is a strictly stronger open refinement of clause (ii), and nothing in the mechanism below reaches it. The type-coding mechanism by itself yields only $H(G(n, W)) = sn \log n + O_W(n \log \log n)$: encoding a latent type for each vertex at metric resolution $1/n$ costs about $s \log n$ nats per vertex and predicts all but on the order of n of the edges, but recording which edges are mispredicted costs order $n \log n$ if done naively, since each such edge must be located by a pair of labels. A two-scale argument, which first codes coarse types and then corrects within coarse classes, lowers the correction cost to $O(n \log \log n)$ and stops there.*

13 A Hessian principle for isolated microcanonical graph phases

Fix a partition

$$[0, 1] = I_1 \sqcup \dots \sqcup I_q, \quad |I_a| = \alpha_a > 0.$$

A coloured graphon is a symmetric measurable $U : [0, 1]^2 \rightarrow [0, 1]$ considered together with this fixed colour partition. Let $p(U) \in [0, 1]^D$ with $D = q(q+1)/2$ denote the block-average vector

$$p_{ab}(U) = \frac{1}{|I_a||I_b|} \int_{I_a \times I_b} U(x, y) dx dy, \quad 1 \leq a \leq b \leq q.$$

A coloured finite graph is a finite simple graph F with a colour $c(v) \in \{1, \dots, q\}$ attached to each vertex, and its coloured homomorphism density in U is

$$t(F, U) = \int_{[0,1]^{V(F)}} \prod_{v \in V(F)} \alpha_{c(v)}^{-1} \mathbf{1}_{I_{c(v)}}(x_v) \prod_{uv \in E(F)} U(x_u, x_v) \prod_{v \in V(F)} dx_v.$$

Fix coloured finite graphs $F_1, \dots, F_k$ and put $\mathbf{t}(U) = (t(F_1, U), \dots, t(F_k, U))$. Define the graphon entropy

$$\mathcal{S}(U) = \frac{1}{2} \int_{[0,1]^2} [-U \log U - (1 - U) \log(1 - U)] dx dy,$$

with $0 \log 0 = 0$, and for a feasible constraint vector $\mathbf{a}$ the contracted entropy

$$\Psi_{\mathbf{a}}(p) = \sup\{\mathcal{S}(U) : \mathbf{t}(U) = \mathbf{a}, p(U) = p\}.$$

The finite graphs are coloured with pinned classes. For each n set $n_a = \lfloor \alpha_a n \rfloor$ and $N_n = \sum_a n_a$, and fix a partition of the vertex set $[N_n]$ into colour classes $V_1, \dots, V_q$ with $|V_a| = \lfloor \alpha_a n \rfloor$ exactly, so that $N_n = n + O(q)$. For a graph G on $[N_n]$ with these colour classes, partition each I_a into n_a consecutive intervals of length $|I_a|/n_a$ assigned to the vertices of V_a , let U_G be the $\{0, 1\}$ -valued coloured graphon equal to 1 on a product of two vertex intervals exactly when the corresponding pair is an edge, and put $\mathbf{t}(G) = \mathbf{t}(U_G)$ and $\widehat{p}(G) = p(U_G)$.

Assume the following regularity hypotheses.

- (H1) The supremum of $\mathcal{S}(U)$ over $\mathbf{t}(U) = \mathbf{a}$ is attained at a unique coloured graphon U_* , up to measure-preserving maps preserving each colour class.
- (H2) Writing $p_* = p(U_*)$, the feasible block-average set $\{p(U) : \mathbf{t}(U) = \mathbf{a}\}$ is, near p_* , a C^3 embedded manifold $M \subset \mathbb{R}^D$.
- (H3) The restriction $\Psi_{\mathbf{a}}|_M$ is C^3 near p_* and has there a unique local maximum.
- (H4) The quadratic form $A = -D_M^2 \Psi_{\mathbf{a}}(p_*)$ is positive definite on the tangent space $T = T_{p_*} M$.
- (H5) [transversally Lipschitz constraint correspondence] There exist $\delta_0 > 0$ and $L < \infty$ such that every coloured graphon U with $\|p(U) - p_*\| \leq \delta_0$ and $\|\mathbf{t}(U) - \mathbf{a}\|_{\infty} = \delta \leq \delta_0$ satisfies $\text{dist}(p(U), M) \leq L\delta$. In particular the conditioning window below confines $\widehat{p}_n$, on the event $\|\widehat{p}_n - p_*\| \leq \delta_0$, to an $o(n^{-1})$ thickening of M .
- (H6) [nonemptiness] For every sufficiently large n the conditioning set $\{G : \|\mathbf{t}(G) - \mathbf{a}\|_{\infty} \leq n^{-1-\eta}\}$ over graphs G on $[N_n]$ with the pinned colour classes is nonempty.

Fix $0 < \eta < 1$, let G_n be uniform on the conditioning set of (H6), let $\widehat{p}_n = \widehat{p}(G_n)$, and let Π_T be orthogonal projection of $\mathbb{R}^D$ onto T .

Conjecture 54 (Hessian principle for an isolated microcanonical phase). *Under hypotheses (H1)–(H6),*

$$n \Pi_T(\widehat{p}_n - \mathbb{E} \widehat{p}_n) \xrightarrow{d} N_T(0, A^{-1}),$$

where $N_T(0, A^{-1})$ denotes the centred Gaussian law on T whose precision form is A .

14 Sharp stability of entropy idempotence on finite abelian groups

Let G be a finite abelian group and μ a probability measure on G . Write $H(\mu)$ for Shannon entropy with natural logarithms, $\mu * \mu$ for the convolution, the law of $X + Y$ with X, Y independent of law μ , and

$$\Delta(\mu) = H(\mu * \mu) - H(\mu)$$

for the convolution entropy defect. If $H \leq G$ is a subgroup and $a \in G$, let u_{a+H} be the uniform measure on the coset $a + H$. Total variation distance is

$$\|\mu - \nu\|_{\text{TV}} = \frac{1}{2} \sum_{x \in G} |\mu(x) - \nu(x)|.$$

Conjecture 55 (Sharp stability of entropy idempotence). *There exists a universal constant $C < \infty$ such that for every finite abelian group G and every probability measure μ on G ,*

$$\inf_{\substack{H \leq G \\ a \in G}} \|\mu - u_{a+H}\|_{\text{TV}}^2 \leq C(H(\mu * \mu) - H(\mu)).$$

Remark 2. *The equality case is verified. With X, Y independent of law μ ,*

$$\Delta(\mu) = H(X + Y) - H(X) = H(X + Y) - H(X + Y | Y) = I(X + Y; Y) \geq 0,$$

where I denotes mutual information. If $\Delta(\mu) = 0$ then $X + Y$ is independent of Y , which forces the translates of μ by the elements of its support to coincide, and hence forces μ to be uniform on a coset of a subgroup. The right-hand side therefore vanishes exactly on the proposed extremizers.

15 Stationary convolution-entropy rigidity

Let K be a finite abelian group and $X = K^{\mathbb{Z}}$ with coordinatewise addition and the left shift σ . Let μ be a σ -invariant ergodic Borel probability measure on X , and write $h_\sigma(\mu)$ for its Kolmogorov–Sinai entropy. Define the translational stabilizer

$$H_\mu = \{h \in K^{\mathbb{Z}} : (\tau_h)_* \mu = \mu\}, \quad \tau_h(x) = x + h,$$

a closed shift-invariant subgroup of $K^{\mathbb{Z}}$, and let $\pi_\mu : K^{\mathbb{Z}} \rightarrow K^{\mathbb{Z}}/H_\mu$ be the quotient homomorphism. Let $\mu * \mu$ be the law of $X + Y$ for independent $X, Y \sim \mu$.

Conjecture 56 (Stationary convolution-entropy rigidity). *For every shift-ergodic μ ,*

$$h_\sigma(\mu * \mu) = h_\sigma(\mu) \iff h_\sigma((\pi_\mu)_* \mu) = 0.$$

Remark 3. *The binary specialization deserves prominence. Take $K = \mathbb{Z}/2\mathbb{Z}$. By Kitchens [35]’ structure theory of group shifts, every closed shift-invariant subgroup of $(\mathbb{Z}/2\mathbb{Z})^{\mathbb{Z}}$ is a group shift of finite type, and every proper one is the solution set of a nontrivial linear recurrence and hence finite. If H_μ is finite the quotient map has finite fibres and $h_\sigma((\pi_\mu)_* \mu) = h_\sigma(\mu)$, while if H_μ is everything then μ is the Haar measure, the uniform Bernoulli process, with entropy $\log 2$. Since $h_\sigma(\mu * \mu) \geq h_\sigma(\mu)$ always, by conditioning on one summand, the conjecture therefore asserts: every ergodic binary process with $0 < h_\sigma(\mu) < \log 2$ gains entropy under independent XOR, the coordinatewise sum modulo 2 with an independent copy.*

Proposition 1 (Easy direction, conditional). Suppose $h_\sigma((\pi_\mu)_*\mu) = 0$. Assume the following two ingredients.

- (a) The Abramov–Rokhlin entropy addition formula holds for the Haar-fibred compact group extension $\pi_\mu : (X, \mu, \sigma) \rightarrow (X/H_\mu, (\pi_\mu)_*\mu, \sigma)$ and for the corresponding extension of $\mu * \mu$, so that each total entropy splits as the quotient entropy plus the common Haar fibre entropy, the conditional measures of μ and of $\mu * \mu$ over the quotient being Haar measures on cosets of H_μ .
- (b) The entropy of a factor of an independent joining is bounded by the sum of the entropies of the factors, so that $(\pi_\mu)_*(\mu * \mu) = ((\pi_\mu)_*\mu) * ((\pi_\mu)_*\mu)$ has entropy at most $2h_\sigma((\pi_\mu)_*\mu) = 0$.

Then $h_\sigma(\mu * \mu) = h_\sigma(\mu)$.

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